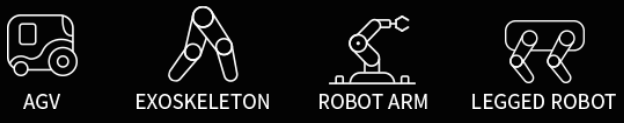
CUBEMARS

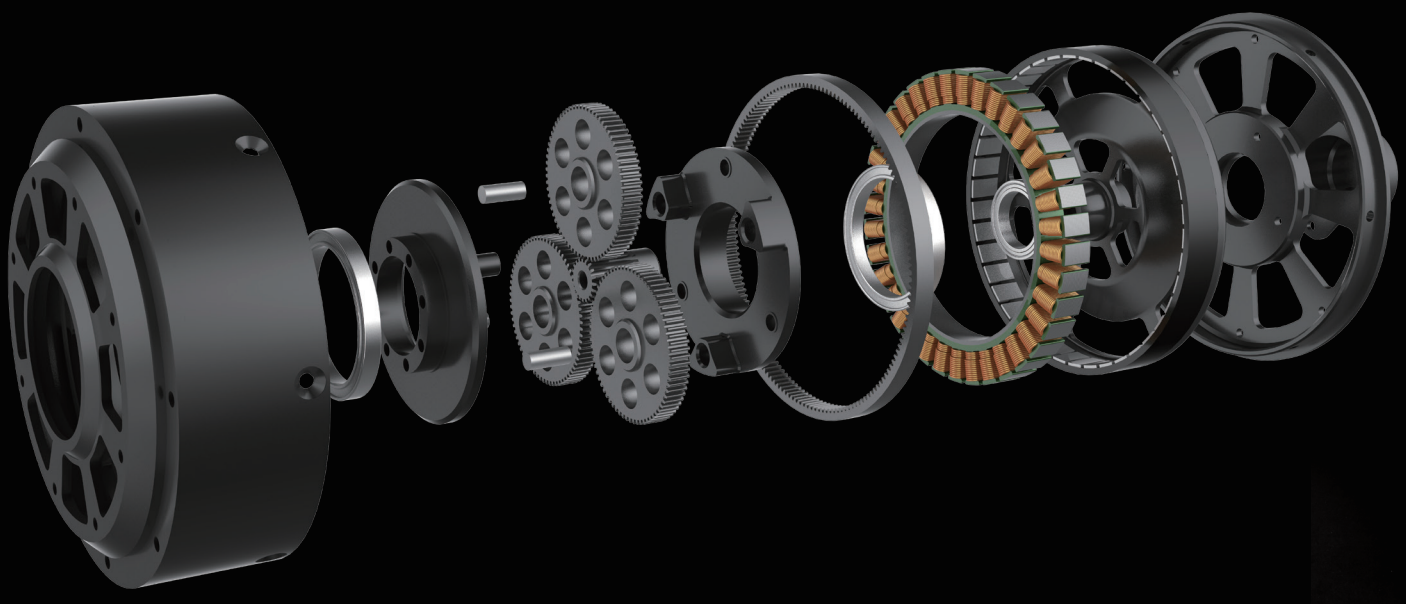
ROBOTIC MOTOR SOLUTION

PRODUCT CATALOG



CUBEMARS

MOTIVATE ADVANCED ROBOTIC SYSTEM



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AKH Series P23-31
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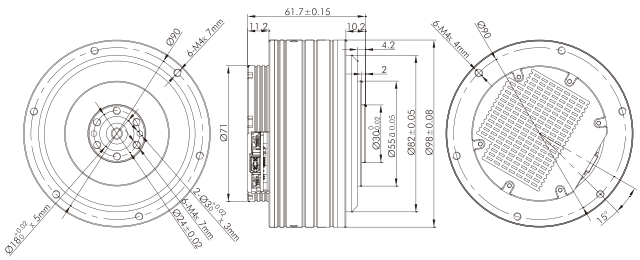
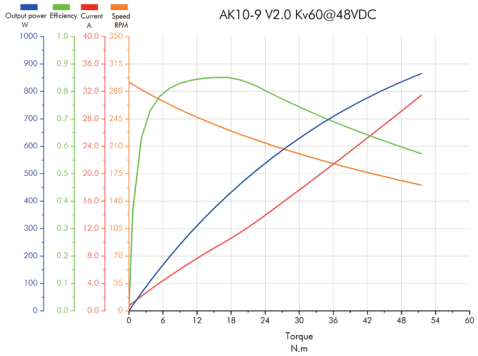
AK10-9 V2.0 KV60, Φ98*61.7mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	9:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.8
Winding Type	Star	Backlash (°)	0.33
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	65
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

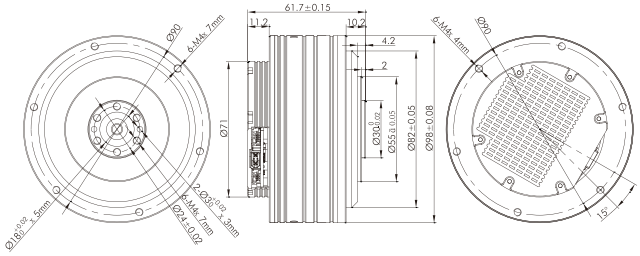
Rated Voltage (V)	24/48	Ke (V/kRPM)	17.2
Rated Torque (N·m)	18	Phase to Phase Resistance (mΩ)	195
Rated Speed (RPM)	109/228	Phase to Phase Inductance (μH)	181
Rated Current (A)	10.6	Inertia (g·cm²)	1002
Peak Torque (N·m)	48	Km (N·m/√W)	0.45
Peak Current (A)	29.8	Mechanical Time Constant (ms)	0.50
Kv (RPM/V)	60	Electrical Time Constant (ms)	0.93
Kt (N·m/A)	0.198	Weight (g)	960
		Maximum Torque Weight Ratio (N·m/kg)	50

CONNECTOR (WITH DRIVER)

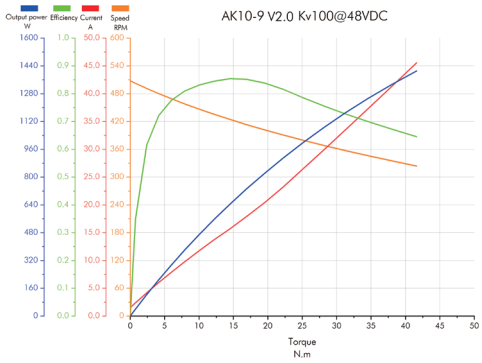
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	Magnetic Encoder
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	15bit
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	2

AK10-9 V2.0 KV100, Φ98*61.7mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	9:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.8
Winding Type	Delta	Backlash (°)	0.33
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	70
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

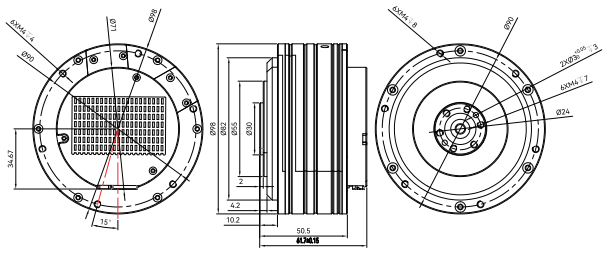
Rated Voltage (V)	24/48	Ke (V/kRPM)	9.9
Rated Torque (N·m)	15	Phase to Phase Resistance (mΩ)	65.5
Rated Speed (RPM)	205/421	Phase to Phase Inductance (μH)	60
Rated Current (A)	16.2	Inertia (g·cm²)	1002
Peak Torque (N·m)	38	Km (N·m/√W)	0.45
Peak Current (A)	41.2	Mechanical Time Constant (ms)	0.51
Kv (RPM/V)	100	Electrical Time Constant (ms)	0.92
Kt (N·m/A)	0.114	Weight (g)	960
		Maximum Torque Weight Ratio (N·m/kg)	39.6

CONNECTOR (WITH DRIVER)

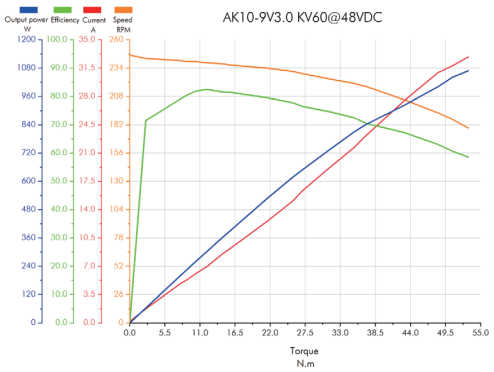
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	Magnetic Encoder
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	15bit
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	2

AK10-9 V3.0 KV60, Φ98*61.7mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	9:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.8
Winding Type	Star	Backlash (°)	0.33
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	70
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

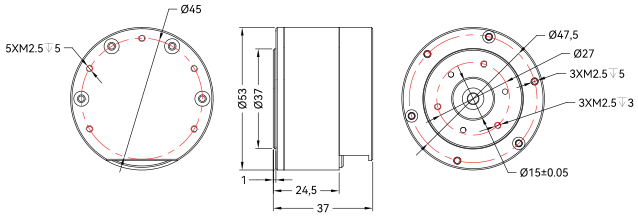
Rated Voltage (V)	48	Ke (V/kRPM)	16.7
Rated Torque (N·m)	18	Phase to Phase Resistance (mΩ)	248
Rated Speed (RPM)	235	Phase to Phase Inductance (μH)	213
Rated Current (A)	10.7	Inertia (g·cm²)	1002
Peak Torque (N·m)	53	Km (N·m/√W)	0.32
Peak Current (A)	31.9	Mechanical Time Constant (ms)	0.5
Kv (RPM/V)	60	Electrical Time Constant (ms)	0.93
Kt (N·m/A)	0.16	Weight (g)	940
		Maximum Torque Weight Ratio (N·m/kg)	86

CONNECTOR (WITH DRIVER)

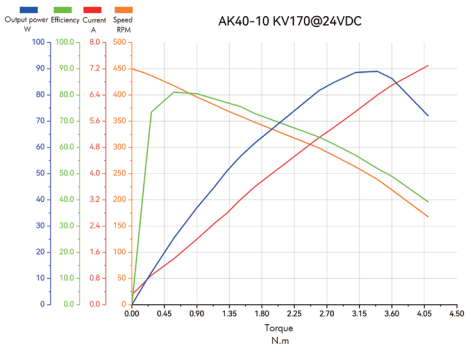
CAN Connector	XT30PW(2+2)-M	Inner Ring Encoder Resolution	21bit
UART Connector	A1257WR-S-3P	Outer Ring Encoder Type	Magnetic Encoder
Power Connector	Magnetic Encoder	Outer Ring Encoder Resolution	15bit
Inner Loop Encoder Type		Number of Encoder	2

AK40-10 KV170, Φ53*37mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



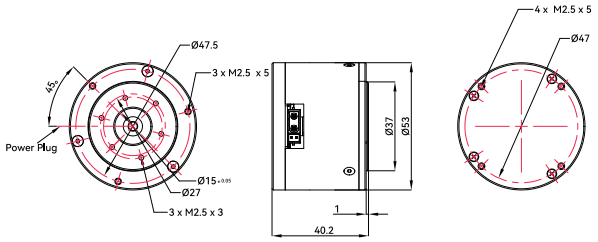
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	C
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10mΩ
Number of Slots	24
Pole Pairs	14
Reduction Ratio	10:1
Back Drive (N·m)	无
Backlash (arcmin)	18
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	50
Basic Dynamic Load Rating Cr (N)	2810
Basic Static Load Rating C_0r (N)	2760
Rated Voltage (V)	24
Rated Torque (N·m)	1.3
Rated Speed (RPM)	370
Rated Current (A)	2.7
Weight (g)	200

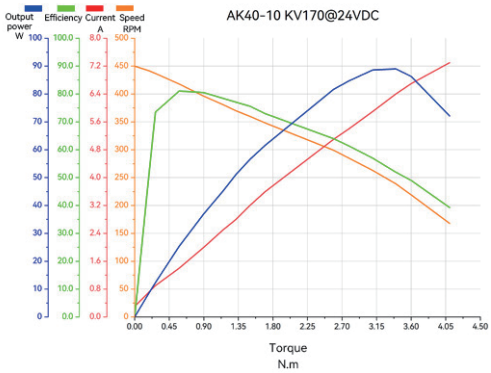
Peak Torque (N·m)	4.1
Peak Current (A)	7.3
No-Load Speed (RPM)	435
Kv (RPM/V)	170
Kt (N·m/A)	0.056
Ke (V/kRPM)	5.88
Phase to Phase Resistance (mΩ)	978
Phase to Phase Inductance (μH)	465
Rotor Inertia (g·cm²)	540
Motor Constant (N·m/√W)	0.0568
Mechanical Time Constant (ms)	2.2
Maximum Torque Density (N·m/kg)	20.5
CAN Connector	A1257WR-S-3P
UART Connector	A1257WR-S-3P
Power Connector	XT30PW-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	14bit
Outer Ring Encoder Type	-
Outer Ring Encoder Resolution	-
Number of Encoder	1

AK40-10 V3.0 KV170, Φ53*40.2mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

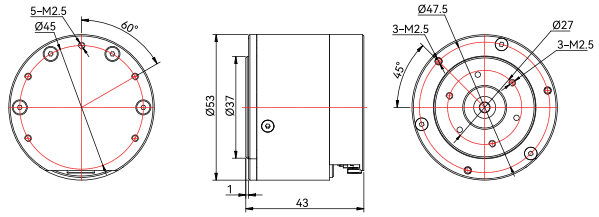
Application	Legged Robot, Exoskeleton, AGV	Number of Slots	24
Driving Way	FOC	Insulation Level	C
Reduction Ratio	10:1	Insulation High-Voltage	500V 5mA/2s
Communication Method	CAN	Insulation Resistance	500V10MΩ
Operation Ambient Temperature	-20℃~50℃	Inner Loop Encoder Type	Magnetic Encoder MT6835
UART Connector	A1257WR-S-3P	Inner Ring Encoder Resolution	16bit
CAN Connector	XT30PW (2+2) -M	Outer Ring Encoder Type	Magnetic Encoder MT6835
Temperature Sensor	NTC MF51B 103F3950	Outer Ring Encoder Resolution	16bit
Pole Pairs	14	Number of Encoder	2
Weight (g)	190		

ELECTRIC PARAMETERS

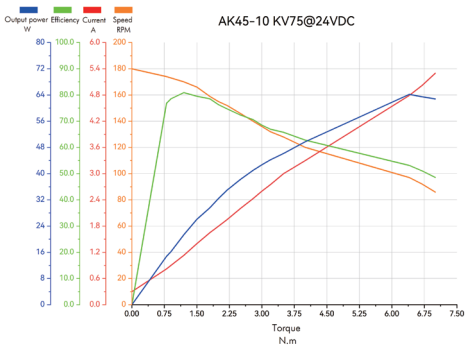
Max Power (W)	50	Noise dB (65cm away the motor)	50
Rated Voltage (V)	24	Back Drive (N·m)	0.06
Rated Torque (N·m)	1.3	Backlash (arcmin)	18
Peak Torque (N·m)	4.1	Rotor Inertia ($g \cdot cm^2$)	97.35
Rated Speed (RPM)	370	Motor Constant (N·m/√W)	0.0568
No-Load Speed (RPM)	435	Kv (RPM/V)	170
Rated Current (A)	2.7	Kt (N·m/A)	0.056
Peak Current (A)	7.3	Ke (V/kRPM)	5.88
Maximum Torque Density (N·m/kg)	20.5	Phase to Phase Resistance (mΩ)	978
Basic Dynamic Load Rating Cr (N)	2810	Phase to Phase Inductance (uH)	465
Basic Static Load Rating C_{0r} (N)	2760		

AK45-10 KV75, Φ53*43mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



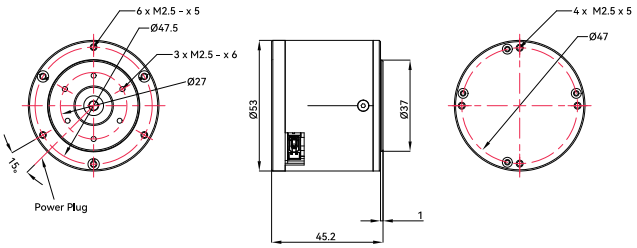
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	C
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10mΩ
Number of Slots	24
Pole Pairs	14
Reduction Ratio	10:1
Back Drive (N·m)	-
Backlash (arcmin)	18
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	60
Basic Dynamic Load Rating Cr (N)	2810
Basic Static Load Rating C_0r (N)	2760
Rated Voltage (V)	24
Rated Torque (N·m)	2.5
Rated Speed (RPM)	150
Rated Current (A)	2.1
Weight (g)	260

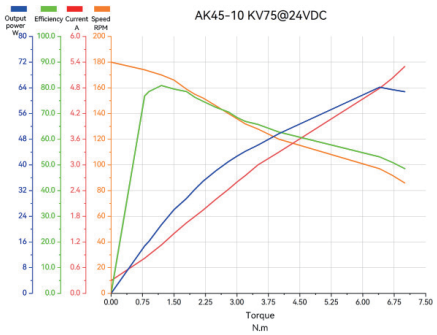
Peak Torque (N·m)	7
Peak Current (A)	5
No-Load Speed (RPM)	180
Kv (RPM/V)	75
Kt (N·m/A)	0.127
Ke (V/kRPM)	13.33
Phase to Phase Resistance (mΩ)	2200
Phase to Phase Inductance (μH)	1330
Rotor Inertia (g·cm²)	89
Motor Constant (N·m/√W)	0.0858
Mechanical Time Constant (ms)	0.6
Maximum Torque Density (N·m/kg)	26.9
CAN Connector	A1257WR-S-3P
UART Connector	A1257WR-S-3P
Power Connector	XT30PW-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	14bit
Outer Ring Encoder Type	-
Outer Ring Encoder Resolution	-
Number of Encoder	1

AK45-10 V3.0 KV75, Φ53*45.2mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

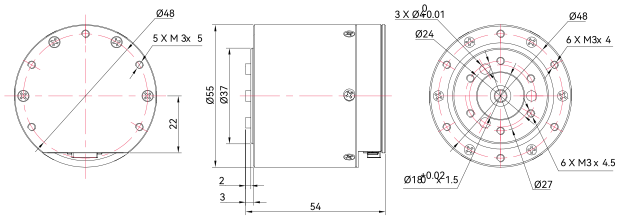
Application	Legged Robot, Exoskeleton, AGV	Number of Slots	24
Driving Way	FOC	Insulation Level	C
Reduction Ratio	10:1	Insulation High-Voltage	500V 5mA/2s
Communication Method	CAN	Insulation Resistance	500V10MΩ
Operation Ambient Temperature	-20℃~50℃	Inner Loop Encoder Type	Magnetic Encoder MT6835
UART Connector	A1257WR-S-3P	Inner Ring Encoder Resolution	16bit
CAN Connector	XT30PW (2+2) -M	Outer Ring Encoder Type	Magnetic Encoder MT6835
Temperature Sensor	NTC MF51B 103F3950	Outer Ring Encoder Resolution	16bit
Pole Pairs	14	Number of Encoder	2
Weight (g)	262		

ELECTRIC PARAMETERS

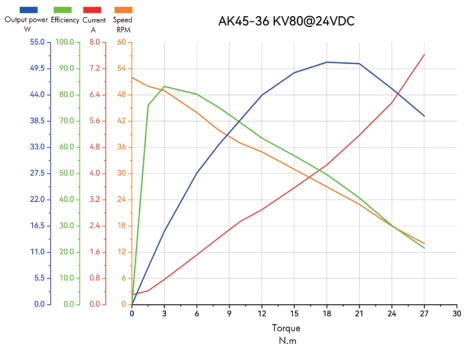
Max Power (W)	39	Noise dB (65cm away the motor)	50
Rated Voltage (V)	24	Back Drive (N·m)	0.1
Rated Torque (N·m)	2.5	Backlash (arcmin)	18
Peak Torque (N·m)	7	Rotor Inertia ($g \cdot cm^2$)	157.33
Rated Speed (RPM)	120	Motor Constant (N·m/√W)	0.0858
No-Load Speed (RPM)	180	Kv (RPM/V)	75
Rated Current (A)	1.9	Kt (N·m/A)	0.127
Peak Current (A)	5	Ke (V/kRPM)	13.33
Maximum Torque Density (N·m/kg)	26.9	Phase to Phase Resistance (mΩ)	2200
Basic Dynamic Load Rating Cr (N)	2810	Phase to Phase Inductance (uH)	1330
Basic Static Load Rating C_0r (N)	2760		

AK45-36 KV80, Φ55*54mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



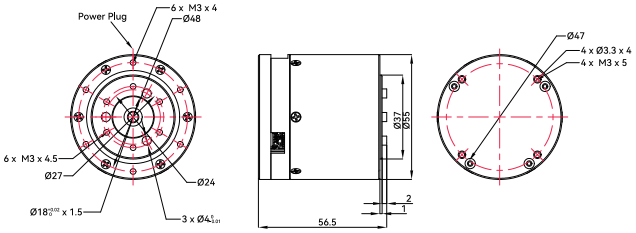
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Level	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Number of Slots	24
Pole Pairs	14
Reduction Ratio	36:1
Back Drive (N·m)	0.8
Backlash (arcmin)	12
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	65
Basic Dynamic Load Rating Cr (N)	2810
Basic Static Load Rating C_0r (N)	2760
Rated Voltage (V)	24
Rated Torque (N·m)	8
Rated Speed (RPM)	40
Rated Current (A)	2
Weight (g)	340

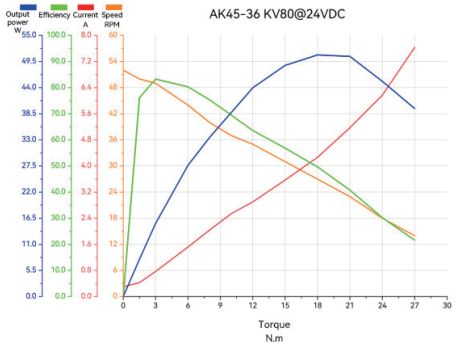
Peak Torque (N·m)	24
Peak Current (A)	6.5
No-Load Speed (RPM)	52
Kv (RPM/V)	80
Kt (N·m/A)	0.11
Ke (V/kRPM)	12.5
Phase to Phase Resistance (mΩ)	1.8
Phase to Phase Inductance (μH)	1.1
Rotor Inertia (g·cm²)	32
Motor Constant (N·m/√W)	0.15
Mechanical Time Constant (ms)	0.83
Maximum Torque Density (N·m/kg)	A1257WR-S-3P
CAN Connector	A1257WR-S-3P
UART Connector	XT30PW-M
Power Connector	Magnetic Encoder
Inner Loop Encoder Type	14bit
Inner Ring Encoder Resolution	-
Outer Ring Encoder Type	-
Outer Ring Encoder Resolution	1
Number of Encoder	33

AK45-36 V3.0 KV80, Φ55*56.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

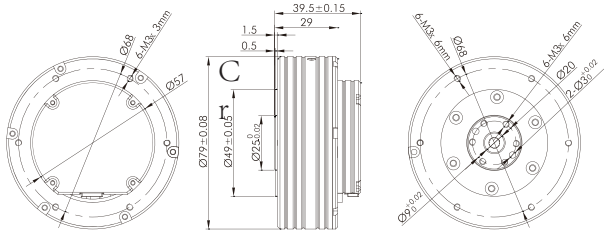
Application	Legged Robot, Exoskeleton, AGV	Number of Slots	24
Driving Way	FOC	Insulation Level	C
Reduction Ratio	36:1	Insulation High-Voltage	500V 5mA/2s
Communication Method	CAN	Insulation Resistance	500V10MΩ
Operation Ambient Temperature	-20℃~50℃	Inner Loop Encoder Type	Magnetic Encoder MT6835
UART Connector	A1257WR-S-3P	Inner Ring Encoder Resolution	16bit
CAN Connector	XT30PW (2+2) -M	Outer Ring Encoder Type	Magnetic Encoder MT6835
Temperature Sensor	NTC MF51B 103F3950	Outer Ring Encoder Resolution	16bit
Pole Pairs	14	Number of Encoder	2
Weight (g)	349		

ELECTRIC PARAMETERS

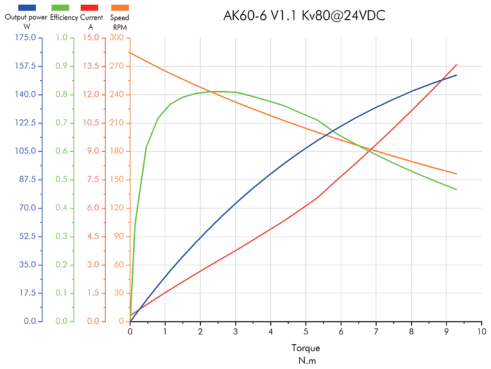
Max Power (W)	33.5	Noise dB (65cm away the motor)	50
Rated Voltage (V)	24	Back Drive (N·m)	0.8
Rated Torque (N·m)	8	Backlash (arcmin)	12
Peak Torque (N·m)	24	Rotor Inertia (g·cm^2)	181.9
Rated Speed (RPM)	40	Motor Constant (N·m/√W)	0.15
No-Load Speed (RPM)	52	Kv (RPM/V)	80
Rated Current (A)	2	Kt (N·m/A)	0.11
Peak Current (A)	6.5	Ke (V/kRPM)	12.5
Maximum Torque Density (N·m/kg)	70.5	Phase to Phase Resistance (mΩ)	1800
Basic Dynamic Load Rating Cr (N)	2810	Phase to Phase Inductance (uH)	1100
Basic Static Load Rating C_0r (N)	2760		

AK60-6 V1.1 KV80, Φ79*39.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	14
Driving Way	FOC	Reduction Ratio	6:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.2
Winding Type	Star	Backlash (°)	0.55
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	500V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	55
Insulation Resistance	500V 10MΩ	Basic Load Rating (dyn. C_r) N	3150
Phase	3	Basic Load Ratings (stat. C_{0r}) N	4500

ELECTRIC PARAMETERS

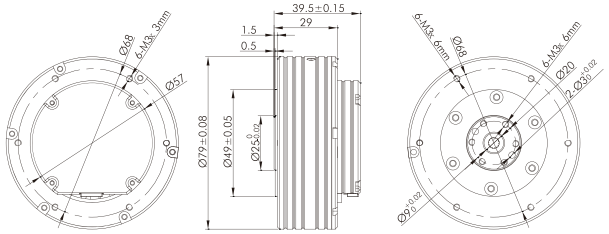
Rated Voltage (V)	24	Ke (V/kRPM)	12.8
Rated Torque (N·m)	3	Phase to Phase Resistance (mΩ)	605
Rated Speed (RPM)	233	Phase to Phase Inductance (μH)	415
Rated Current (A)	3.8	Inertia (g·cm²)	243.5
Peak Torque (N·m)	9	Km (N·m/√W)	0.17
Peak Current (A)	13.1	Mechanical Time Constant (ms)	0.81
Kv (RPM/V)	80	Electrical Time Constant (ms)	0.69
Kt (N·m/A)	0.135	Weight (g)	368
		Maximum Torque Weight Ratio (N·m/kg)	24.46

CONNECTOR (WITH DRIVER)

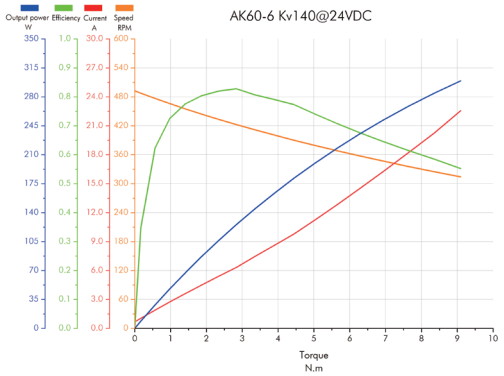
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

AK60-6 V1.1 KV140, Φ79*39.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	14
Driving Way	FOC	Reduction Ratio	6:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.2
Winding Type	Delta	Backlash (°)	0.55
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	500V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	60
Insulation Resistance	500V 10MΩ	Basic Load Rating (dyn. C_r) N	3150
Phase	3	Basic Load Ratings (stat. C_{0r}) N	4500

ELECTRIC PARAMETERS

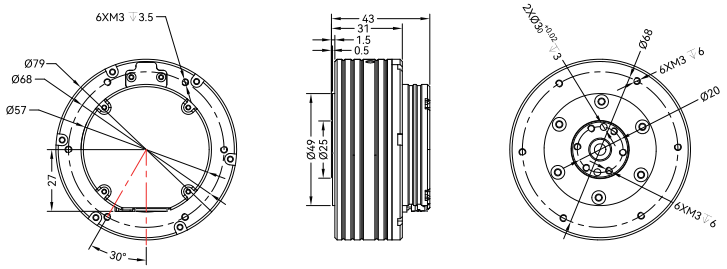
Rated Voltage (V)	24	Ke (V/kRPM)	7.5
Rated Torque (N·m)	3	Phase to Phase Resistance (mΩ)	202
Rated Speed (RPM)	420	Phase to Phase Inductance (μH)	138
Rated Current (A)	6.5	Inertia (g·cm²)	243.5
Peak Torque (N·m)	9	Km (N·m/√W)	0.17
Peak Current (A)	22.7	Mechanical Time Constant (ms)	0.81
Kv (RPM/V)	140	Electrical Time Constant (ms)	0.68
Kt (N·m/A)	0.078	Weight (g)	368
		Maximum Torque Weight Ratio (N·m/kg)	24.46

CONNECTOR (WITH DRIVER)

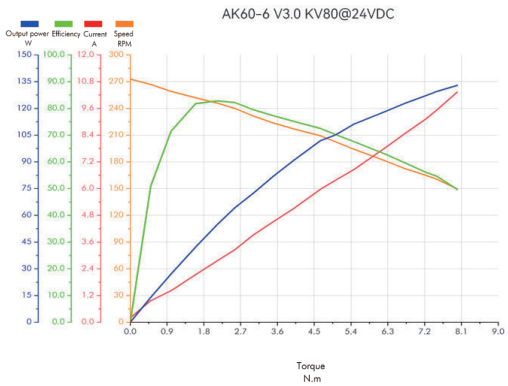
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

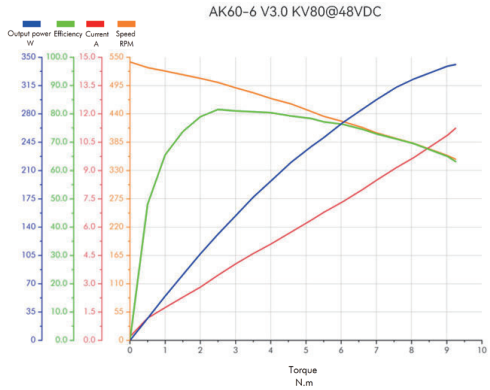
AK60-6 V3.0 KV80, Φ79*43mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION





SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	14
Driving Way	FOC	Reduction Ratio	6:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.2
Winding Type	Star	Backlash (°)	0.55
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	500V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	55
Insulation Resistance	500V 10MΩ	Basic Load Rating (dyn. C_r) N	3150
Phase	3	Basic Load Ratings (stat. C_{0r}) N	4500

ELECTRIC PARAMETERS

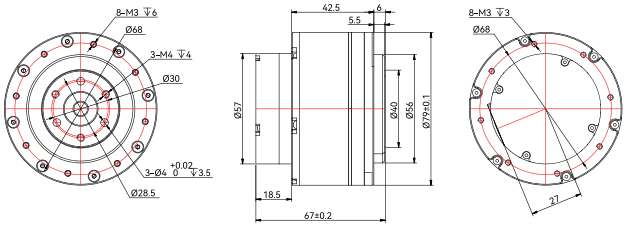
Rated Voltage (V)	24/48	Ke (V/kRPM)	12.5
Rated Torque (N·m)	3	Phase to Phase Resistance (mΩ)	595
Rated Speed (RPM)	233/490	Phase to Phase Inductance (μH)	676
Rated Current (A)	3.8	Inertia (g·cm²)	243.5
Peak Torque (N·m)	9	Km (N·m/√W)	0.15
Peak Current (A)	0.3/11.2	Mechanical Time Constant (ms)	0.81
Kv (RPM/V)	80	Electrical Time Constant (ms)	0.69
Kt (N·m/A)	0.135	Weight (g)	380
		Maximum Torque Weight Ratio (N·m/kg)	23.68

CONNECTOR (WITH DRIVER)

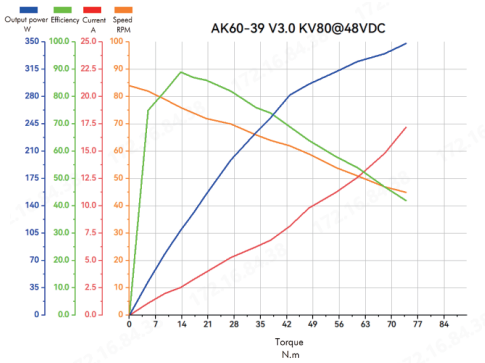
POWER+CAN Connector	XT30PW(2+2)-M	Inner Ring Encoder Resolution	16bit
UART Connector	A1257WR-S-3P	Number of Encoder	1
Inner Loop Encoder Type	Magnetic Encoder		

AK60-39 V3.0 KV80, Φ79*48.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



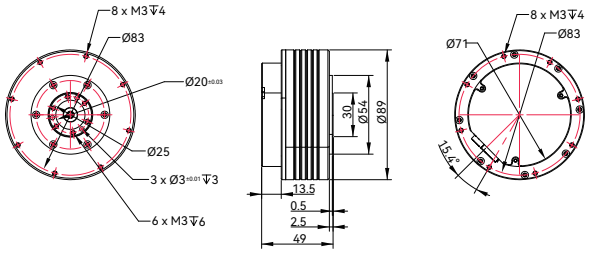
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Level	C
Insulation High-Voltage	AC 500V 5mA/2s
Insulation Resistance	DC 500V 10mΩ
Number of Slots	24
Pole Pairs	14
Reduction Ratio	39:1
Back Drive (N·m)	1.4
Backlash (arcmin)	15
Temperature Sensor	NTC MF51B 103F3950
Noise dB Away the Motor	60
Basic Dynamic Load Rating Cr (N)	2000
Basic Static Load Rating C_0r (N)	2520
Rated Voltage (V)	48
Rated Torque (N·m)	24
Rated Speed (RPM)	70
Rated Current (A)	4.5

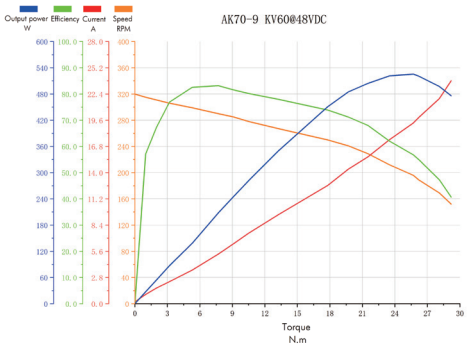
Peak Torque (N·m)	72
Peak Current (A)	17
No-Load Speed (RPM)	98
Kv (RPM/V)	80
Kt (N·m/A)	0.12
Ke (V/kRPM)	12.5
Phase to Phase Resistance (mΩ)	600
Phase to Phase Inductance (μH)	670
Rotor Inertia (g·cm²)	305
Motor Constant (N·m/√W)	0.15
Mechanical Time Constant (ms)	2.39
Maximum Torque Density (N·m/kg)	96
CAN Connector	A1257WR-S-3P
UART Connector	XT30PW-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	21bit
Outer Ring Encoder type	Magnetic Encoder
Outer Ring Encoder Resolution	21bit
Number of Encoder	2
Weight (g)	750

AK70-9 V3.0 KV60, Φ89*49mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



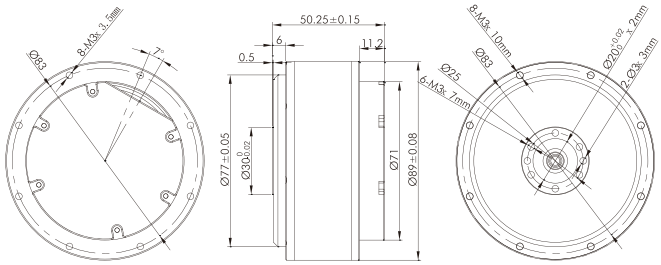
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	C
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Number of Slots	24
Pole Pairs	21
Reduction Ratio	9:1
Back Drive (N·m)	0.8
Backlash (arcmin)	18
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	50
Basic Dynamic Load Rating Cr (N)	1890
Basic Static Load Rating C_0r (N)	2140
Rated Voltage (V)	48
Rated Torque (N·m)	8.5
Rated Speed (RPM)	260
Rated Current (A)	6.25
Weight (g)	540

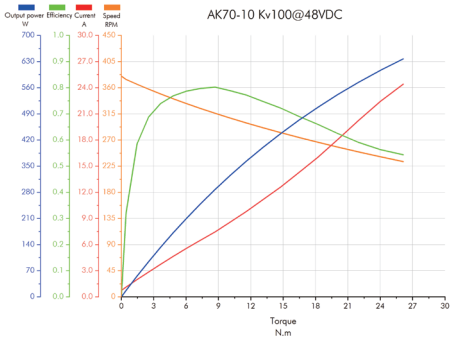
Peak Torque (N·m)	29.2
Peak Current (A)	23.8
No-Load Speed (RPM)	320
Kv (RPM/V)	60
Kt (N·m/A)	0.159
Ke (V/kRPM)	16.7
Phase to Phase Resistance (mΩ)	475
Phase to Phase Inductance (μH)	408
Rotor Inertia (g·cm²)	450
Motor Constant (N·m/√W)	0.23
Mechanical Time Constant (ms)	1.7
Maximum Torque Density (N·m/kg)	54
CAN Connector	A1257WR-S-4P
UART Connector	A1257WR-S-3P
Power Connector	XT30PW-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	14bit
Outer Ring Encoder Type	Magnetic Encoder
Outer Ring Encoder Resolution	15bit
Number of Encoder	2

AK70-10 KV100, Φ89*50.25mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	10:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.48
Winding Type	Delta	Backlash (°)	0.12
Insulation Class	H	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	58
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

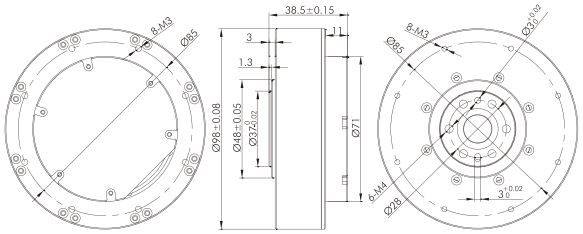
Rated Voltage (V)	24/48	Ke (V/kRPM)	11.2
Rated Torque (N·m)	8.3	Phase to Phase Resistance (mΩ)	272
Rated Speed (RPM)	148/310	Phase to Phase Inductance (μH)	113
Rated Current (A)	7.2	Inertia (g·cm²)	414
Peak Torque (N·m)	24.8	Km (N·m/√W)	0.24
Peak Current (A)	23.2	Mechanical Time Constant (ms)	0.74
Kv (RPM/V)	100	Electrical Time Constant (ms)	0.42
Kt (N·m/A)	0.123	Weight (g)	621
		Maximum Torque Weight Ratio (N·m/kg)	47.6

CONNECTOR (WITH DRIVER)

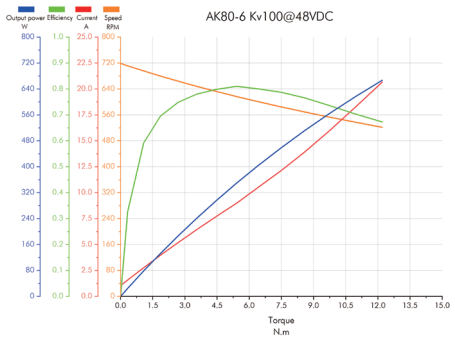
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

AK80-6 KV100, Φ98*38.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	6:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.36
Winding Type	Delta	Backlash (°)	0.22
Insulation Class	H	Temperature Sensor	None
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	55
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2760
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2810

ELECTRIC PARAMETERS

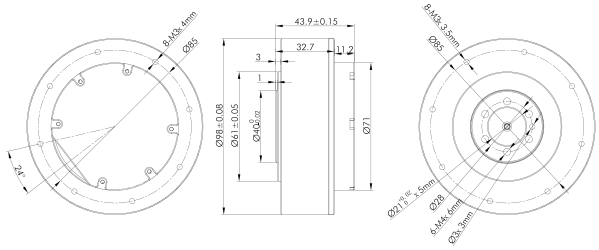
Rated Voltage (V)	48	Ke (V/kRPM)	110.5
Rated Torque (N·m)	6	Phase to Phase Resistance (mΩ)	170
Rated Speed (RPM)	603	Phase to Phase Inductance (μH)	57
Rated Current (A)	9.7	Inertia (g·cm²)	607
Peak Torque (N·m)	12	Km (N·m/√W)	0.25
Peak Current (A)	20	Mechanical Time Constant (ms)	0.94
Kv (RPM/V)	100	Electrical Time Constant (ms)	0.34
Kt (N·m/A)	0.105	Weight (g)	485
		Maximum Torque Weight Ratio (N·m/kg)	24

CONNECTOR (WITH DRIVER)

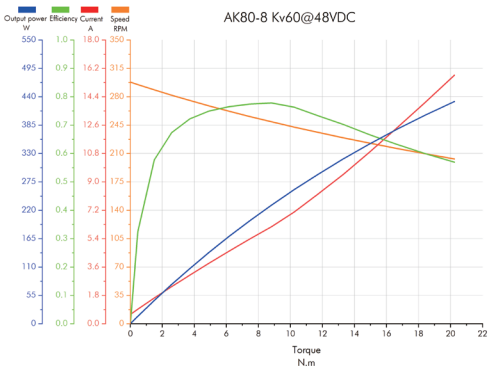
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

AK80-8 KV60, Φ98*43.9mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	8:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.75
Winding Type	Star	Backlash (°)	0.38
Insulation Class	C	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	50
Insulation Resistance	1000V 10MvΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

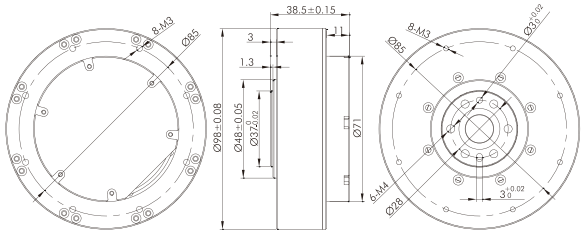
Rated Voltage (V)	48	Ke (V/kRPM)	19.8
Rated Torque (N·m)	10	Phase to Phase Resistance (mΩ)	430
Rated Speed (RPM)	243	Phase to Phase Inductance (μH)	214
Rated Current (A)	6.9	Inertia (g·cm²)	608.6
Peak Torque (N·m)	25	Km (N·m/√W)	0.30
Peak Current (A)	21	Mechanical Time Constant (ms)	0.66
Kv (RPM/V)	60	Electrical Time Constant (ms)	0.5
Kt (N·m/A)	0.199	Weight (g)	570
		Maximum Torque Weight Ratio (N·m/kg)	35

CONNECTOR (WITH DRIVER)

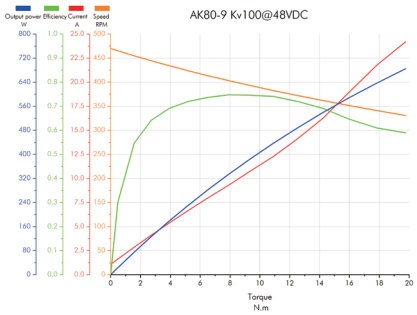
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	Magnetic Encoder
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	15bit
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	2

AK80-9 KV100, Φ98*38.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	9:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	0.51
Winding Type	Delta	Backlash (°)	0.19
Insulation Class	H	Temperature Sensor	None
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	55
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2760
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2810

ELECTRIC PARAMETERS

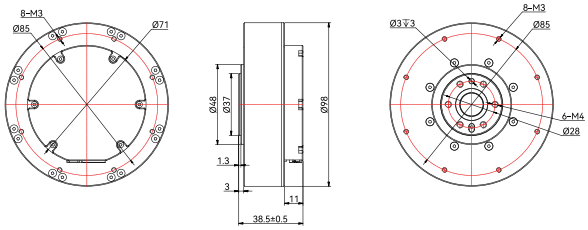
Rated Voltage (V)	48	Ke (V/kRPM)	10.5
Rated Torque (N·m)	9	Phase to Phase Resistance (mΩ)	170
Rated Speed (RPM)	390	Phase to Phase Inductance (μH)	57
Rated Current (A)	10.3	Inertia (g·cm^2)	607
Peak Torque (N·m	18	Km (N·m/√W)	0.25
Peak Current (A)	22.3	Mechanical Time Constant (ms)	0.94
Kv (RPM/V)	100	Electrical Time Constant (ms)	0.34
Kt (N·m/A)	0.105	Weight (g)	485
		Maximum Torque Weight Ratio (N·m/kg)	37

CONNECTOR (WITH DRIVER)

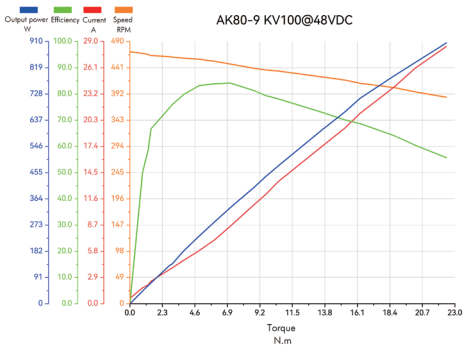
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

AK80-9 V3.0 KV100, Φ98*38.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



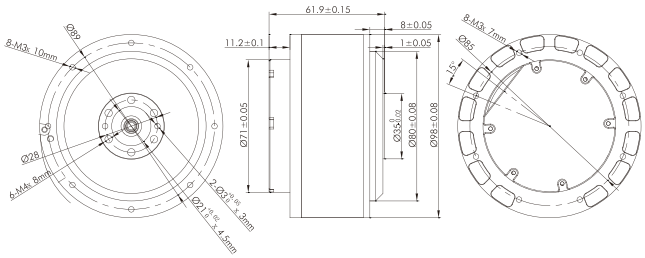
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	C
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Number of Slots	36
Pole Pairs	21
Reduction Ratio	9:1
Back Drive (N·m)	0.51
Backlash (arcmin)	15
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	60
Basic Dynamic Load Rating Cr (N)	2760
Basic Static Load Rating C_0r (N)	2810
Rated Voltage (V)	48
Rated Torque (N·m)	9
Rated Speed (RPM)	390
Rated Current (A)	12
Weight (g)	490

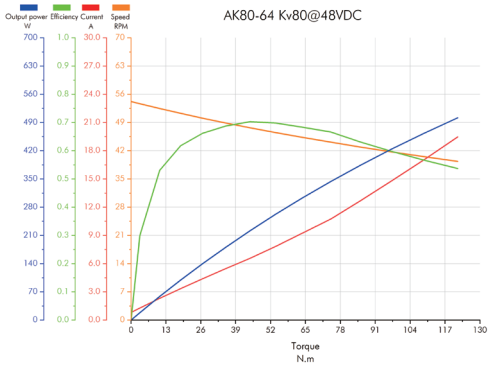
Peak Torque (N·m)	22
Peak Current (A)	28
No-Load Speed (RPM)	570
Kv (RPM/V)	100
Kt (N·m/A)	0.095
Ke (V/kRPM)	10
Phase to Phase Resistance (mΩ)	160
Phase to Phase Inductance (μH)	116
Rotor Inertia (g·cm²)	579
Motor Constant (N·m/√W)	0.2387
Mechanical Time Constant (ms)	0.725
Maximum Torque Density (N·m/kg)	44.9
CAN Connector	-
UART Connector	A1257WR-S-3P
Power Connector	XT30PW-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	16bit
Outer Ring Encoder Type	-
Outer Ring Encoder Resolution	-
Number of Encoder	1

AK80-64 KV80, Φ98*61.9mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV	Pole Pairs	21
Driving Way	FOC	Reduction Ratio	64:1
Operation Ambient Temperature	-20℃~50℃	Back Drive (N·m)	4.7
Winding Type	Delta	Backlash (°)	0.18
Insulation Class	H	Temperature Sensor	NTC MF51B 103F3950
Insulation High-Voltage	1000V 5mA/2s	Noise dB 65cm Away the Motor (Environment 45dB@1200 RPM)	60
Insulation Resistance	1000V 10MΩ	Basic Load Rating (dyn. C_r) N	2000
Phase	3	Basic Load Ratings (stat. C_{0r}) N	2520

ELECTRIC PARAMETERS

Rated Voltage (V)	24/48	Ke (V/kRPM)	13.7
Rated Torque (N·m)	48	Phase to Phase Resistance (mΩ)	220
Rated Speed (RPM)	23/48	Phase to Phase Inductance (μH)	133.5
Rated Current (A)	7	Inertia (g·cm²)	564.5
Peak Torque (N·m)	120	Km (N·m/√W)	0.29
Peak Current (A)	19	Mechanical Time Constant (ms)	0.67
Kv (RPM/V)	80	Electrical Time Constant (ms)	0.61
Kt (N·m/A)	0.136	Weight (g)	850
		Maximum Torque Weight Ratio (N·m/kg)	141.2

CONNECTOR (WITH DRIVER)

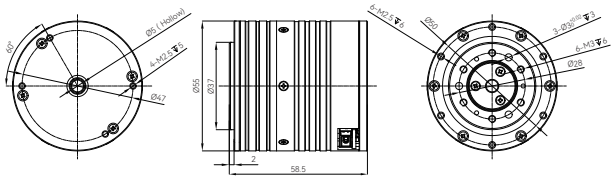
CAN Connector	A1257WR-S-4P	Inner Ring Encoder Resolution	14bit
UART Connector	A1257WR-S-4P	Outer Ring Encoder Type	-
Power Connector	XT30PW-M	Outer Ring Encoder Resolution	-
Inner Loop Encoder Type	Magnetic Encoder	Number of Encoder	1

AKH Series Hollow Shaft Modules

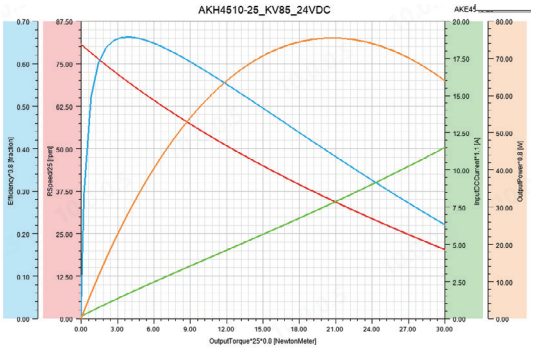
AKH40-25 24
AKH50-25 25
AKH60-25 26
AKH70-16 27
AKH70-48 29
AKH90-16 30
AKH90-48 31

AKH40-25 KV85, Φ55*58.5mm

PRODUCT DRAWING



Module External Characteristic Curve



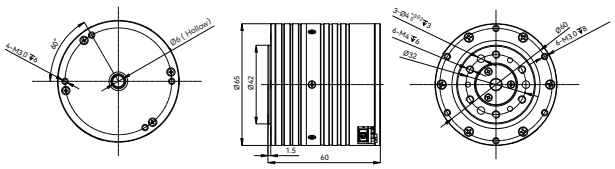
ANALOG DATA

Voltage (V)	24	Module Rated Torque (N·m)	7
Reduction Ratio	25	Module Rated Speed (RPM)	60
Reducer Efficiency (%)	80	Module Rated Phase Current (Arms)	2.4
Stator Core	4510	Module Rated Bus Current (A)	2.9
Number of Slots and Poles	24N28P	Module Torque Constant (N·m/Arms)	2.9
Module Max No-Load Speed (RPM)	81	Module Peak Torque (N·m)	21
Line Resistance (mΩ)	1.7	Module Peak Phase Current (Arms)	7.5
Line inductance (mH)	0.6		

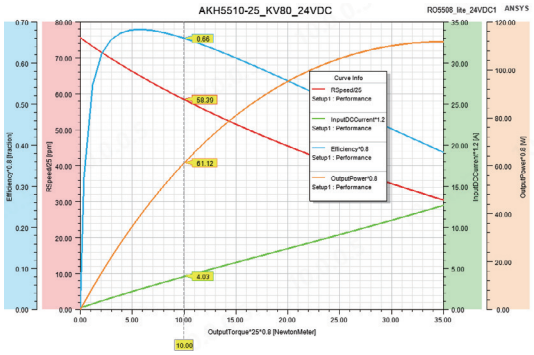
The above data is based on software simulation and is for model selection reference only.
The actual measured prototype shall prevail.

AKH50-25 KV80, Φ65*60mm

PRODUCT DRAWING



Module External Characteristic Curve



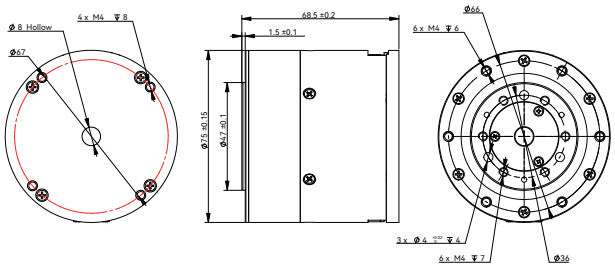
模拟数据

Voltage (V)	24	Module Rated Torque (N·m)	10
Reduction Ratio	25	Module Rated Speed (RPM)	55
Reducer Efficiency (%)	80	Module Rated Phase Current (Arms)	3.45
Stator Core	5510	Module Rated Bus Current (A)	4
Number of Slots and Poles	24N28P	Module Torque Constant (N·m/Arms)	2.9
Module Max No-Load Speed (RPM)	76	Module Peak Torque (N·m)	30
Line Resistance (mΩ)	0.95	Module Peak Phase Current (Arms)	10.4
Line inductance (mH)	0.6		

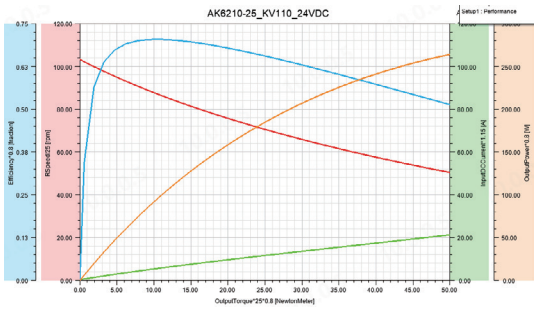
The above data is based on software simulation and is for model selection reference only.
The actual measured prototype shall prevail.

AKH60-25 KV110, Φ75*68.5mm

PRODUCT DRAWING



Module External Characteristic Curve



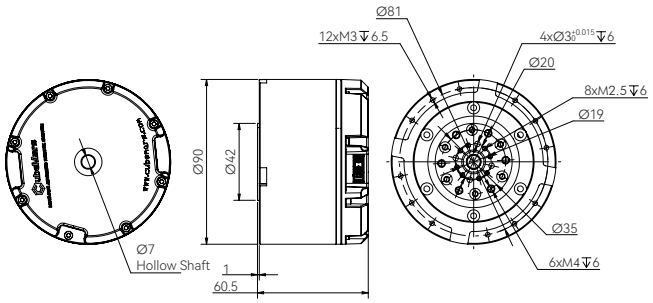
模拟数据

Voltage (V)	24	Module Rated Torque (N·m)	13.5
Reduction Ratio	25	Module Rated Speed (RPM)	80
Stator Core	4510	Module Rated Phase Current (Arms)	6.3
Number of Slots and Poles	24N28P	Module Rated Bus Current (A)	6.9
Module Max No-Load Speed (RPM)	81	Module Torque Constant (N·m/Arms)	2.14
Line Resistance (mΩ)	1.7	Module Peak Torque (N·m)	41
Line inductance (mH)	0.6	Module Peak Phase Current (Arms)	21

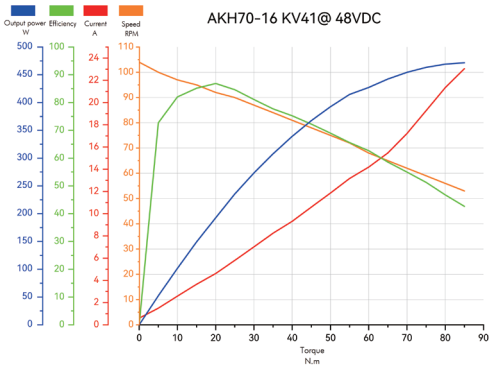
The above data is based on software simulation and is for model selection reference only.
The actual measured prototype shall prevail.

AKH70-16 KV41, Φ90*60.5mm

PRODUCT DRAWING



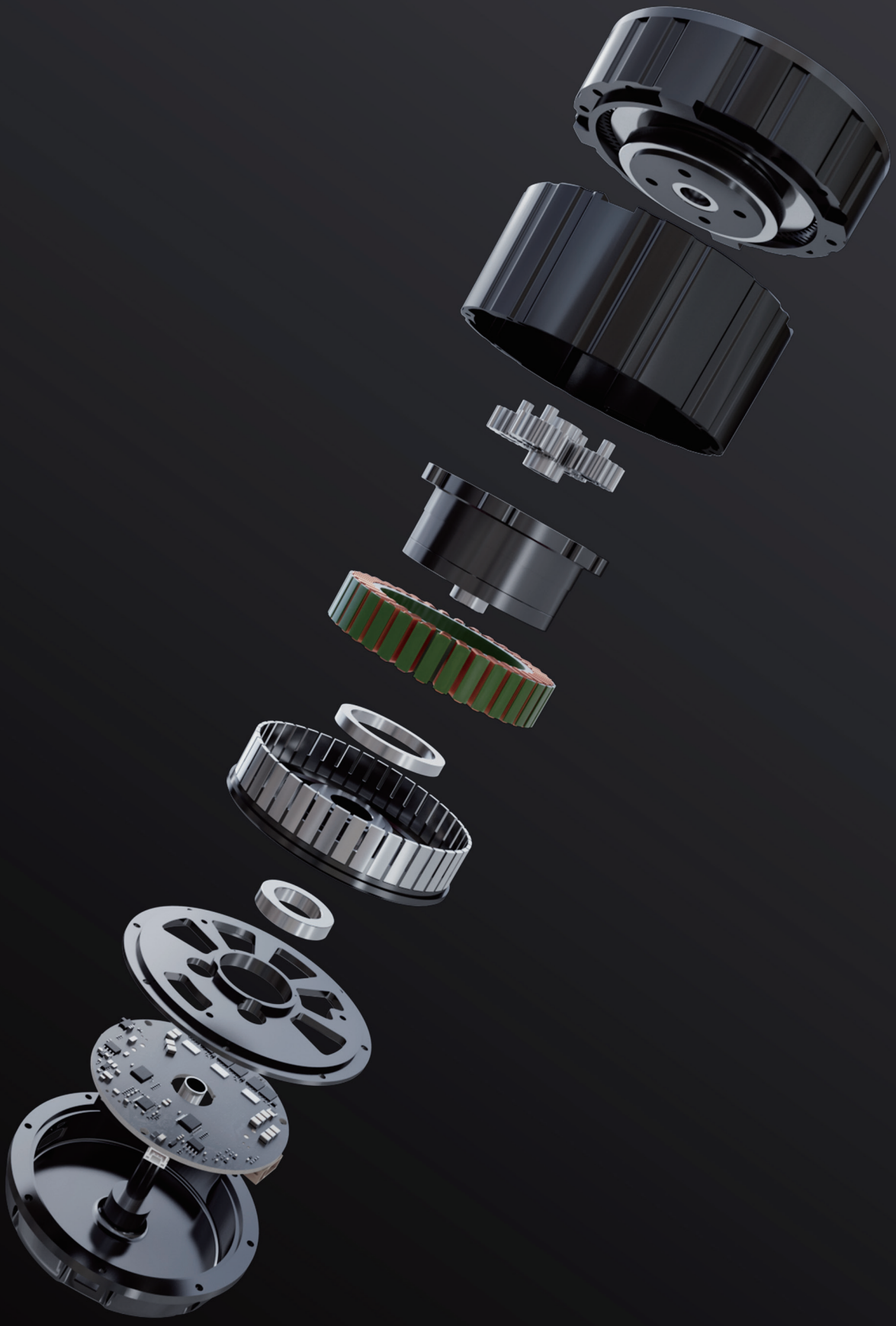
ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

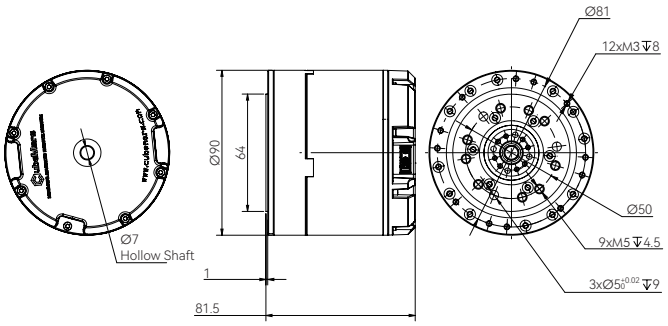
Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10mΩ
Number of Slots	36
Pole Pairs	21
Reduction Ratio	16:1
Back Drive (N·m)	0.78
Backlash (arcmin)	12
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	65
Basic Dynamic Load Rating Cr (N)	6350
Basic Static Load Rating C_0r (N)	5550
Rated Voltage (V)	48
Rated Torque (N·m)	26
Rated Speed (RPM)	90
Rated Current (A)	6.5
Weight (g)	879

Peak Torque (N·m)	85
Peak Current (A)	23
No-Load Speed (RPM)	105
Kv (RPM/V)	41
Kt (N·m/A)	0.23
Ke (V/kRPM)	2.4
Phase to Phase Resistance (mΩ)	0.56
Phase to Phase Inductance (μH)	0.6
Rotor Inertia (g·cm²)	1170
Motor Constant (N·m/√W)	0.31
Mechanical Time Constant (ms)	2.28
Maximum Torque Density (N·m/kg)	88.74
CAN Connector	XT30PW(2+2)-M
UART Connector	A1257WR-S-3P
Power Connector	XT30PW(2+2)-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	21bit
Outer Ring Encoder Type	Magnetic Encoder
Outer Ring Encoder Resolution	21bit
Number of Encoder	2

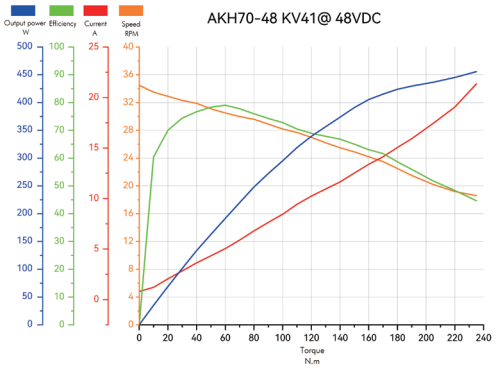


AKH70-48 KV41, Φ90*81.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



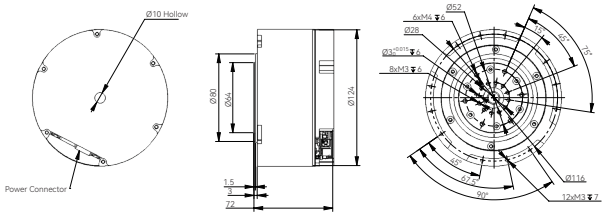
SPECIFICATIONS

Application	Legged Robot, Exoskeleton, AGV
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Level	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10mΩ
Number of Slots	36
Pole Pairs	21
Reduction Ratio	48:1
Back Drive (N·m)	2.22
Backlash (arcmin)	12
Temperature Sensor	NTC MF51B 103F3950
Noise dB (65cm away the motor)	65
Basic Dynamic Load Rating Cr (N)	5680
Basic Static Load Rating C_0r (N)	8680
Rated Voltage (V)	48
Rated Torque (N·m)	74
Rated Speed (RPM)	28
Rated Current (A)	6
Weight (g)	1396

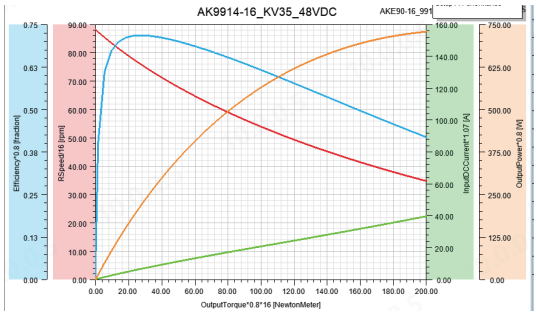
Peak Torque (N·m)	222
Peak Current (A)	21
No-Load Speed (RPM)	35
Kv (RPM/V)	41
Kt (N·m/A)	0.23
Ke (V/kRPM)	2.4
Phase to Phase Resistance (mΩ)	0.56
Phase to Phase Inductance (μH)	0.6
Rotor Inertia (g·cm²)	1170
Motor Constant (N·m/√W)	0.31
Mechanical Time Constant (ms)	2.28
Maximum Torque Density (N·m/kg)	159
CAN Connector	XT30PW(2+2)-M
UART Connector	A1257WR-S-3P
Power Connector	XT30PW(2+2)-M
Inner Loop Encoder Type	Magnetic Encoder
Inner Ring Encoder Resolution	21bit
Outer Ring Encoder Type	Magnetic Encoder
Outer Ring Encoder Resolution	21bit
Number of Encoder	2

AKH90-16 KV35, Φ124*72mm

PRODUCT DRAWING



Module External Characteristic Curve



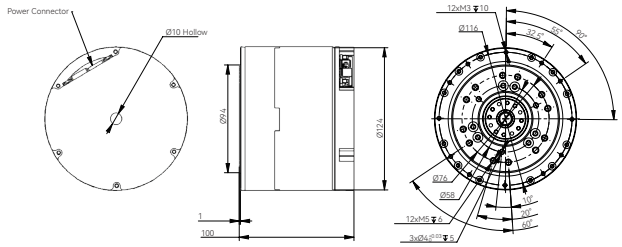
模拟数据

Voltage (V)	48	Line Resistance (mΩ)	0.31
Number of Slots and Poles	36N42P	Line inductance (mH)	0.4
Stator Core	9914	Module Rated Torque (N·m)	50
Reduction Ratio	16	Module Rated Speed (RPM)	65
Module Outer Diameter (mm)	-	Module Rated Phase Current (Arms)	10.7
Estimated Module Height (mm)	-	Module Torque Constant (N·m/Arms)	4.67
Estimated Module Weight (mm)	- (Motor electromagnetic weight 416g)	Module Peak Torque (N·m)	155
		Module Peak Phase Current (Arms)	36.7
Module Max No-Load Speed (RPM)	88		

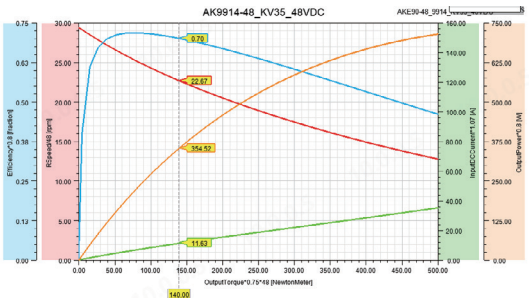
The above data is based on software simulation and is for model selection reference only.
The actual measured prototype shall prevail.

AKH90-48 KV35, Φ124*100mm

PRODUCT DRAWING



Module External Characteristic Curve



模拟数据

Voltage (V)	48	Line Resistance (mΩ)	0.31
Number of Slots and Poles	36N42P	Line inductance (mH)	0.4
Stator Core	9914	Module Rated Torque (N·m)	140
Reduction Ratio	48	Module Rated Speed (RPM)	22
Reducer Efficiency (%)	75	Module Rated Phase Current (Arms)	10.7
Module Outer Diameter (mm)	-	Module Torque Constant (N·m/Arms)	14
Estimated Module Height (mm)	-	Module Peak Torque (N·m)	436
Estimated Module Weight (mm)	- (Motor electromagnetic weight 416g)	Module Peak Phase Current (Arms)	36.7
Module Max No-Load Speed (RPM)	29		

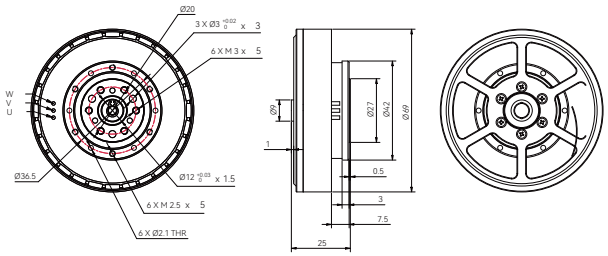
The above data is based on software simulation and is for model selection reference only.
The actual measured prototype shall prevail.

AKE Series Quasi Direct Drive Motors

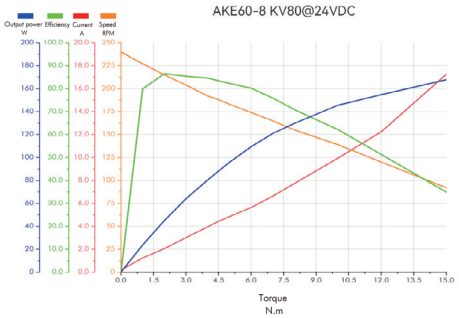
AKE60-8 33
AKE80-8 34
AKE90-8 35

AKE60-8 KV80, Φ69*25mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION

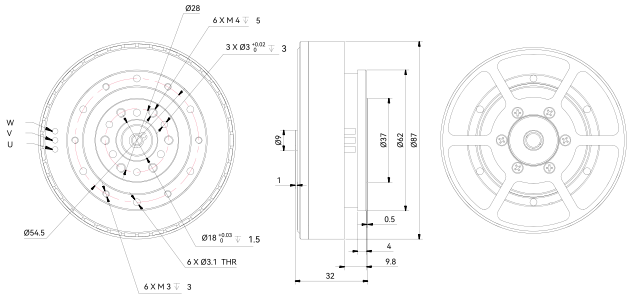


SPECIFICATIONS

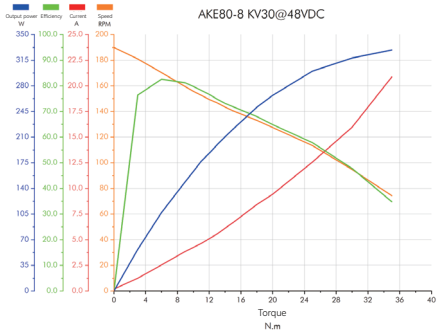
Application	Exoskeleton	Peak Torque (N·m)	12.5
Driving Way	FOC	Peak Current (A)	12
Operation Ambient Temperature (℃)	-20℃~50℃	Kv (RPM/V)	80
Winding Type	Star	Kt (N·m/A)	0.13
Insulation Class	F	Ke (V/kRPM)	12.5
Insulation High-Voltage (V)	500	Phase to Phase Resistance (mΩ)	577
Insulation Resistance (MΩ)	10	Phase to Phase Inductance (μH)	704
Pole Pairs	14	Inertia (g·cm²)	647.39
Arcmin	9	Km (N·m/√W)	0.15
Rated Voltage (V)	24	Mechanical Time Constant (ms)	1.2
No-Load Speed (RPM)	240	Electrical Time Constant (ms)	1.7
Rated Torque (N·m)	5	Weight (g)	260
Rated Speed (RPM)	180	Maximum Torque Weight Ratio (N·m/kg)	46
Rated Current (A)	4.8	Reduction Ratio	8:1

AKE80-8 KV30, ϕ87*32mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION

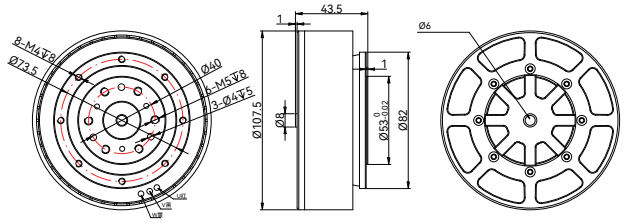


SPECIFICATIONS

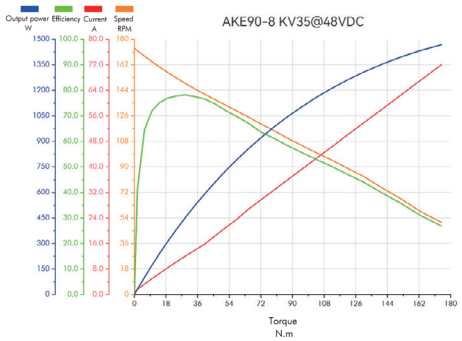
Application	Exoskeleton	Peak Torque (N·m)	30
Driving Way	FOC	Peak Current (A)	12
Operation Ambient Temperature (℃)	-20℃~50℃	Kv (RPM/V)	30
Winding Type	Star	Kt (N·m/A)	0.32
Insulation Class	F	Ke (V/kRPM)	33
Insulation High-Voltage (V)	500	Phase to Phase Resistance (mΩ)	870
Insulation Resistance (MΩ)	10	Phase to Phase Inductance (μH)	990
Pole Pairs	21	Inertia (g·cm²)	143
Arcmin	9	Km (N·m/√W)	0.34
Rated Voltage (V)	48	Mechanical Time Constant (ms)	2.3
No-Load Speed (RPM)	195	Electrical Time Constant (ms)	1.13
Rated Torque (N·m)	12	Weight (g)	570
Rated Speed (RPM)	150	Maximum Torque Weight Ratio (N·m/kg)	52
Rated Current (A)	4.8	Reduction Ratio	8:1

AKE90-8 KV35, Φ107.5*43.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Exoskeleton	Peak Torque (N·m)	170
Driving Way	FOC	Peak Current (A)	72
Operation Ambient Temperature (℃)	-20℃~50℃	Kv (RPM/V)	35
Winding Type	Delta	Kt (N·m/A)	0.272
Insulation Class	H	Ke (V/kRPM)	28.5
Insulation High-Voltage (V)	1000	Phase to Phase Resistance (mΩ)	164
Insulation Resistance (MΩ)	10	Phase to Phase Inductance (μH)	235
Pole Pairs	21	Inertia (g·cm²)	3377.08
Arcmin	9	Km (N·m/√W)	0.67372
Rated Voltage (V)	48	Mechanical Time Constant (ms)	2.18
No-Load Speed (RPM)	210	Electrical Time Constant (ms)	1.4329
Rated Torque (N·m)	55	Weight (g)	1400
Rated Speed (RPM)	120	Maximum Torque Weight Ratio (N·m/kg)	121.4
Rated Current (A)	21	Reduction Ratio	8:1

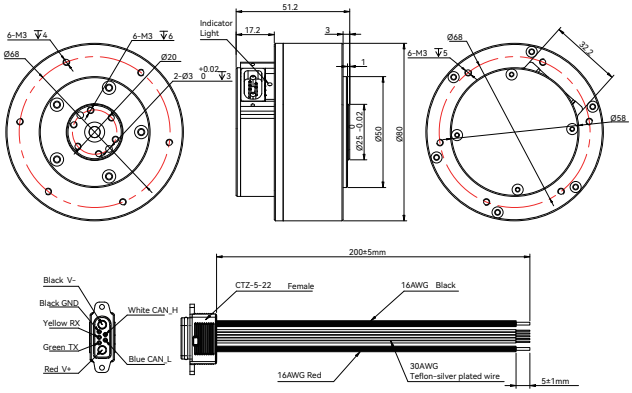
AKA Series Robotic Actuation Module

AKA60-6 37

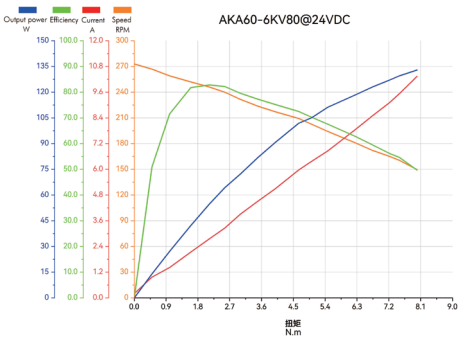
AKA10-9 38

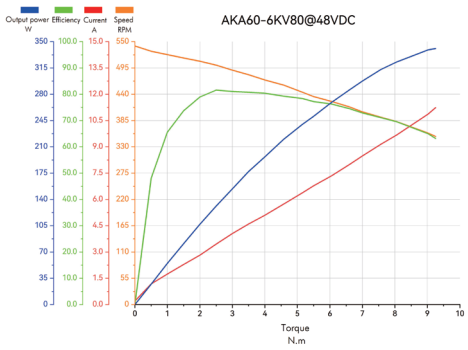
AKA60-6 KV80, Φ80*51.2mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



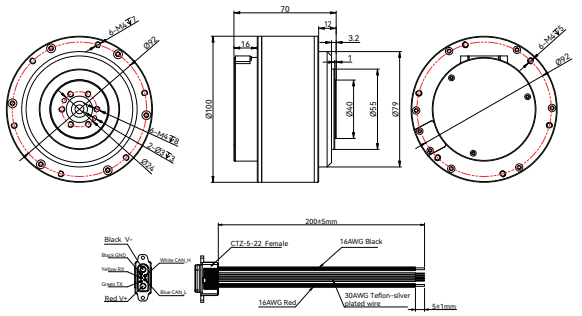


SPECIFICATIONS

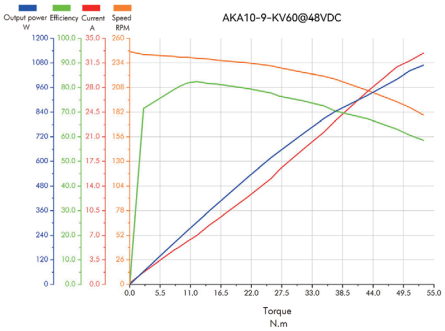
Application	WheeledRobot, AGV	Rated Current (A)	4
Driving Way	FOC	Peak Torque (N·m)	9
Operation Ambient Temperature (℃)	-20℃~50℃	Peak Current (A)	11.2
Winding Type	Star	Kv (RPM/V)	80
Insulation Class	H	Kt (N·m/A)	0.11937
Insulation High-Voltage (V)	500	Ke (V/kRPM)	0.0125
Insulation Resistance (MΩ)	10	Phase to Phase Resistance (mΩ)	595
Pole Pairs	14	Phase to Phase Inductance (μH)	675
Arcmin	9	Inertia (g·cm²)	331.91
Rated Voltage (V)	24/48	Km (N·m/√W)	0.1541
No-Load Speed (RPM)	320/640	Mechanical Time Constant (ms)	2.5
Rated Torque (N·m)	3	Electrical Time Constant (ms)	1.13
Rated Speed (RPM)	200/400	Weight (g)	460

AKA10-9 KV60, Φ100*70mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION

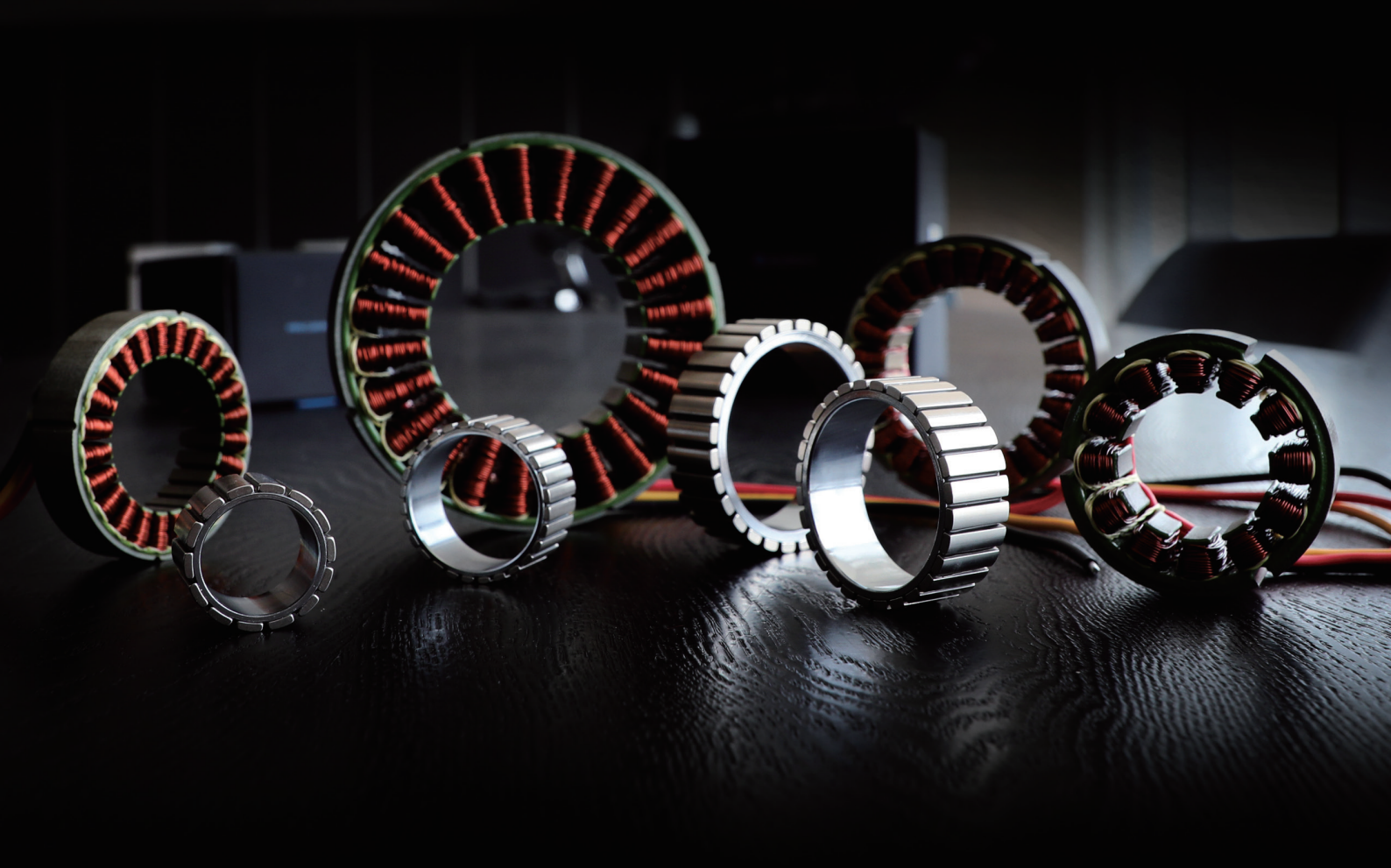


SPECIFICATIONS

Application	WheeledRobot, AGV	Rated Current (A)	10.6
Driving Way	FOC	Peak Torque (N·m)	53
Operation Ambient Temperature (℃)	-20℃~50℃	Peak Current (A)	32
Winding Type	Star	Kv (RPM/V)	60
Insulation Class	C	Kt (N·m/A)	0.16
Insulation High-Voltage (V)	1000	Ke (V/kRPM)	0.0167
Insulation Resistance (MΩ)	10	Phase to Phase Resistance (mΩ)	248
Pole Pairs	14	Phase to Phase Inductance (μH)	235
Arcmin	9	Inertia (g·cm^2)	1002
Rated Voltage (V)	48	Km (N·m/√W)	0.32
No-Load Speed (RPM)	320	Mechanical Time Constant (ms)	1.90
Rated Torque (N·m)	18	Electrical Time Constant (ms)	0.94
Rated Speed (RPM)	109	Weight (g)	1060

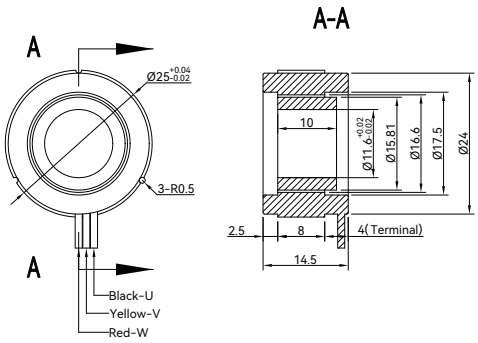
RI Series Frameless Inrunner Torque Motor

RI20 40
RI30 41
RI50 42
RI60 43
RI70 44
RI80 V2.0 45
RI100 46

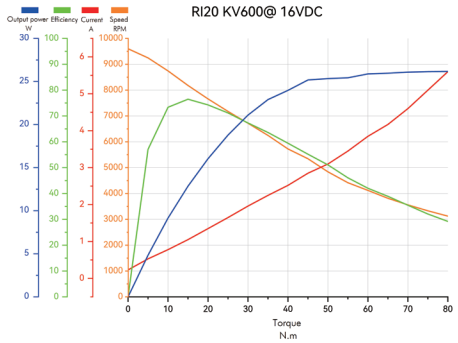


RI20 KV600, Φ25*14.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	H
Driving Way	FOC	Insulation High-Voltage	500V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	500V 10MΩ
Winding Type	-	Phase	-
		Pole Pairs	7

ELECTRIC PARAMETERS

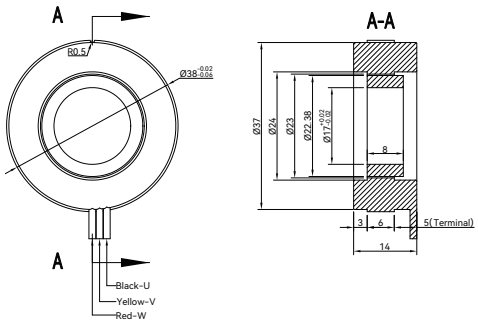
Rated Voltage (V)	16	Ke (V/kRPM)	0.0016
No-Load Speed (RPM)	9600	Phase to Phase Resistance (mΩ)	1890
Rated Torque (N·m)	0.024	Phase to Phase Inductance (μH)	450
Rated Speed (RPM)	7452	Inertia (g·cm²)	1.38
Rated Current (A)	1.47	Km (N·m/√W)	0.0115
Peak Torque (N·m)	0.072	Mechanical Time Constant (ms)	1.03
Peak Current (A)	4.4	Electrical Time Constant (ms)	0.24
Kv (RPM/V)	600	Weight (g)	20
Kt (N·m/A)	0.01592	Maximum Torque Weight Ratio (N·m/kg)	4.85

CONNECTOR

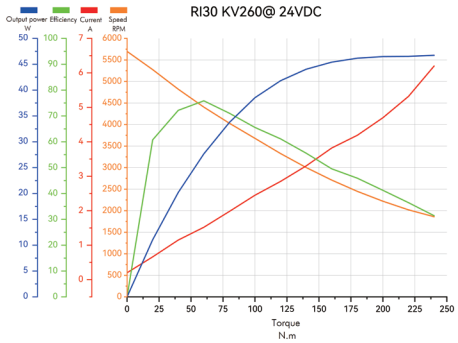
U	Black	Hv	-
V	Yellow	Hw	-
W	Red	VCC GND	-
Hu	-	Correspondence	-

RI30 KV260, Φ38*14mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	H
Driving Way	FOC	Insulation High-Voltage	500V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	500V 10MΩ
Winding Type	-	Phase	-
		Pole Pairs	7

ELECTRIC PARAMETERS

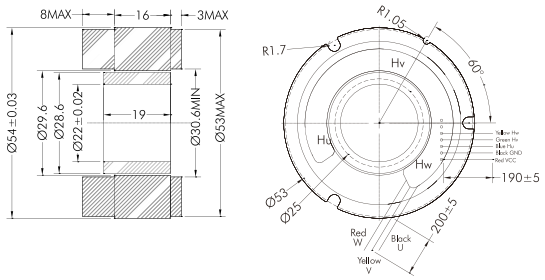
Rated Voltage (V)	24	Ke (V/kRPM)	0.0038
No-Load Speed (RPM)	6240	Phase to Phase Resistance (mΩ)	2200
Rated Torque (N·m)	0.076	Phase to Phase Inductance (μH)	1410
Rated Speed (RPM)	4110	Inertia (g·cm²)	4.23
Rated Current (A)	1.98	Km (N·m/√W)	0.0247
Peak Torque (N·m)	0.23	Mechanical Time Constant (ms)	0.69
Peak Current (A)	7.72	Electrical Time Constant (ms)	0.64
Kv (RPM/V)	260	Weight (g)	40
Kt (N·m/A)	0.03673	Maximum Torque Weight Ratio (N·m/kg)	2.14

CONNECTOR

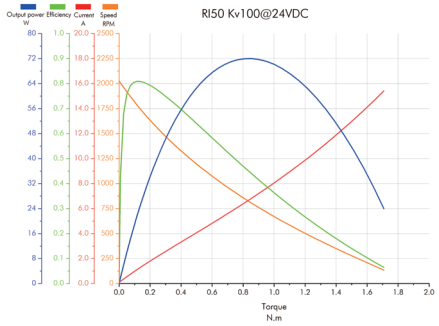
U	Black	Hv	-
V	Yellow	Hw	-
W	Red	VCC GND	-
Hu	-	Correspondence	-

RI50 KV100, Φ54*27mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	C
Driving Way	FOC	Insulation High-Voltage	500V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	500V 10MΩ
Winding Type	Star	Phase	3
		Pole Pairs	7

ELECTRIC PARAMETERS

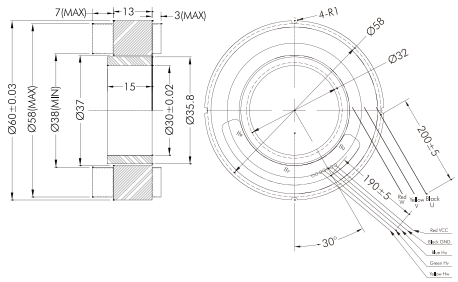
Rated Voltage (V)	24/36/48	Ke (V/kRPM)	11.41
No-Load Speed (RPM)	2004/3006/4008	Phase to Phase Resistance (mΩ)	1420
Rated Torque (N·m)	0.58	Phase to Phase Inductance (μH)	1500
Rated Speed (RPM)	1090/1860/2600	Inertia (g·cm²)	22.8
Rated Current (A)	4.8	Km (N·m/√W)	0.1007
Peak Torque (N·m)	1.67	Mechanical Time Constant (ms)	0.22
Peak Current (A)	14.8	Electrical Time Constant (ms)	1.06
Kv (RPM/V)	100	Weight (g)	180.8
Kt (N·m/A)	0.120	Maximum Torque Weight Ratio (N·m/kg)	9.24

CONNECTOR

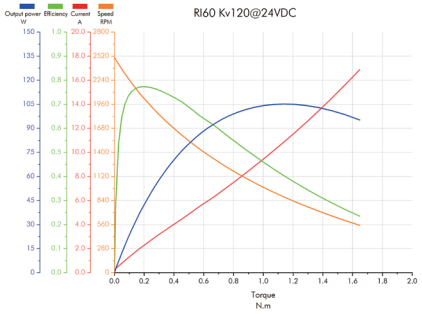
U	Black+18#Silicone Wire	Hv	Green+30#Silicone Wire
V	Yellow+18#Silicone Wire	Hw	Yellow+30#Silicone Wire
W	Red+18#Silicone Wire	VCC GND	Red+30#Silicone Wire Black+30#Silicone Wire
Hu	Blue+30#Silicone Wire	Correspondence	Hu-U Hv-V Hw-W

RI60 KV120, Φ60*23mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	C
Driving Way	FOC	Insulation High-Voltage	500V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	500V 10MΩ
Winding Type	Delta	Phase	3
		Pole Pairs	14

ELECTRIC PARAMETERS

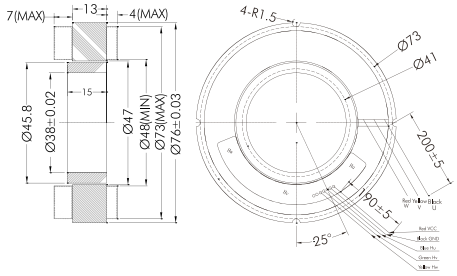
Rated Voltage (V)	24/36/48	Ke (V/kRPM)	9.03
No-Load Speed (RPM)	2532/3798/5064	Phase to Phase Resistance (mΩ)	900
Rated Torque (N·m)	0.57	Phase to Phase Inductance (μH)	877.5
Rated Speed (RPM)	1440/2320/3190	Inertia (g·cm²)	33.05
Rated Current (A)	5.6	Km (N·m/√W)	0.1054
Peak Torque (N·m)	1.63	Mechanical Time Constant (ms)	0.30
Peak Current (A)	16.8	Electrical Time Constant (ms)	0.98
Kv (RPM/V)	120	Weight (g)	155.9
Kt (N·m/A)	0.100	Maximum Torque Weight Ratio (N·m/kg)	10.46

CONNECTOR

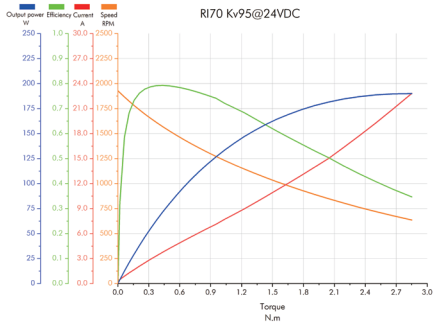
U	Black+18#Silicone Wire	Hv	Green+30#Silicone Wire
V	Yellow+18#Silicone Wire	Hw	Yellow+30#Silicone Wire
W	Red+18#Silicone Wire	VCC GND	Red+30#Silicone Wire Black+30#Silicone Wire
Hu	Blue+30#Silicone Wire	Correspondence	Hu-U Hv-V Hw-W

RI70 KV95, Φ76*24mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	C
Driving Way	FOC	Insulation High-Voltage	500V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	500V 10MΩ
Winding Type	Delta	Phase	3
		Pole Pairs	14

ELECTRIC PARAMETERS

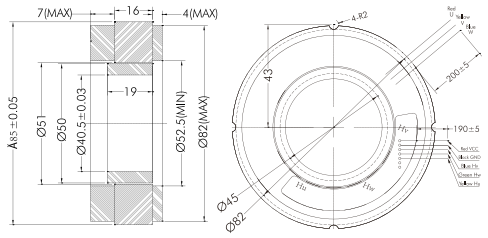
Rated Voltage (V)	24/36/48	Ke (V/kRPM)	11.69
No-Load Speed (RPM)	1956/2934/3912	Phase to Phase Resistance (mΩ)	418
Rated Torque (N·m)	0.94	Phase to Phase Inductance (μH)	622.7
Rated Speed (RPM)	1270/1990/2710	Inertia (g·cm²)	92.15
Rated Current (A)	7.1	Km (N·m/√W)	0.2011
Peak Torque (N·m)	2.68	Mechanical Time Constant (ms)	0.23
Peak Current (A)	21	Electrical Time Constant (ms)	1.49
Kv (RPM/V)	95	Weight (g)	270.4
Kt (N·m/A)	0.130	Maximum Torque Weight Ratio (N·m/kg)	9.91

CONNECTOR

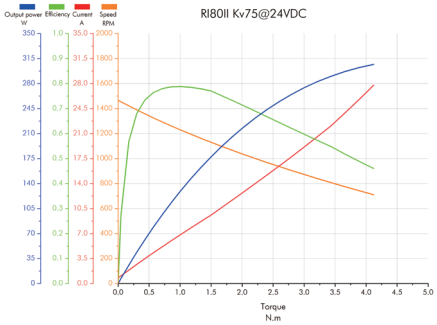
U	Black+16#Silicone Wire	Hv	Green+30#Silicone Wire
V	Yellow+16#Silicone Wire	Hw	Yellow+30#Silicone Wire
W	Red+16#Silicone Wire	VCC GND	Red+30#Silicone Wire Black+30#Silicone Wire
Hu	Blue+30#Silicone Wire		
		Correspondence	Hu-U Hv-V Hw-W
		CCW Correspondence	eg: Hu-U Hv-V Hw-W

RI80 V2.0 KV75, Φ85*27mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	C
Driving Way	FOC	Insulation High-Voltage	1000V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	1000V 10MΩ
Winding Type	Delta	Phase	3
		Pole Pairs	8

ELECTRIC PARAMETERS

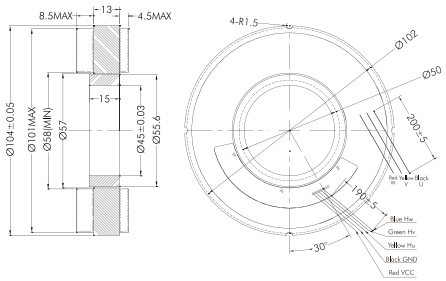
Rated Voltage (V)	24/36/48	Ke (V/kRPM)	15.5
No-Load Speed (RPM)	1480/2220/2960	Phase to Phase Resistance (mΩ)	330
Rated Torque (N·m)	1.45	Phase to Phase Inductance (μH)	510
Rated Speed (RPM)	1135/1785/2430	Inertia (g·cm²)	212.49
Rated Current (A)	9.4	Km (N·m/√W)	0.2698
Peak Torque (N·m)	4.1	Mechanical Time Constant (ms)	0.29
Peak Current (A)	27.6	Electrical Time Constant (ms)	1.55
Kv (RPM/V)	75	Weight (g)	411
Kt (N·m/A)	0.155	Maximum Torque Weight Ratio (N·m/kg)	9.3

CONNECTOR

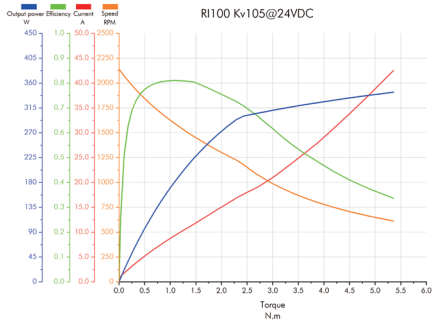
U	Black+16#Silicone Wire	Hv	Green+30#Silicone Wire
V	Yellow+16#Silicone Wire	Hw	Yellow+30#Silicone Wire
W	Red+16#Silicone Wire	VCC GND	Red+30#Silicone Wire Black+30#Silicone Wire
Hu	Blue+30#Silicone Wire	Correspondence	Hu-U Hv-V Hw-W

RI100 KV105, Φ104*26mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton	Insulation Class	C
Driving Way	FOC	Insulation High-Voltage	1000V 5mA/2s
Operation Ambient Temperature	-20℃~50℃	Insulation Resistance	1000V 10MΩ
Winding Type	Delta	Phase	3
		Pole Pairs	14

ELECTRIC PARAMETERS

Rated Voltage (V)	24/36/48	Ke (V/kRPM)	10.47
No-Load Speed (RPM)	2184/3276/4368	Phase to Phase Resistance (mΩ)	126
Rated Torque (N·m)	1.76	Phase to Phase Inductance (μH)	366.7
Rated Speed (RPM)	1370/2100/2825	Inertia (g·cm²)	215.5
Rated Current (A)	13.6	Km (N·m/√W)	0.3634
Peak Torque (N·m)	4.95	Mechanical Time Constant (ms)	0.16
Peak Current (A)	38.6	Electrical Time Constant (ms)	2.91
Kv (RPM/V)	105	Weight (g)	500
Kt (N·m/A)	0.129	Maximum Torque Weight Ratio (N·m/kg)	9.9

CONNECTOR

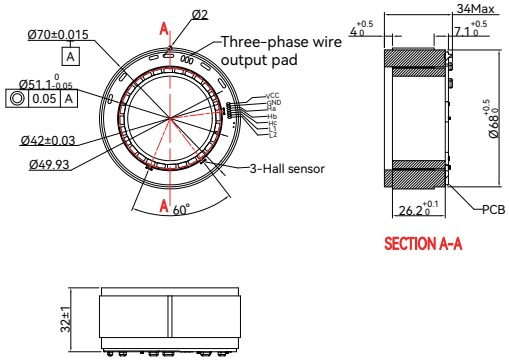
U	Black+16#Silicone Wire	Hv	Green+30#Silicone Wire
V	Yellow+16#Silicone Wire	Hw	Yellow+30#Silicone Wire
W	Red+16#Silicone Wire	VCC GND	Red+30#Silicone Wire Black+30#Silicone Wire
Hu	Blue+30#Silicone Wire	Correspondence	Hu-U Hv-V Hw-W

RI-PH Series Frameless Inrunner Torque Motor

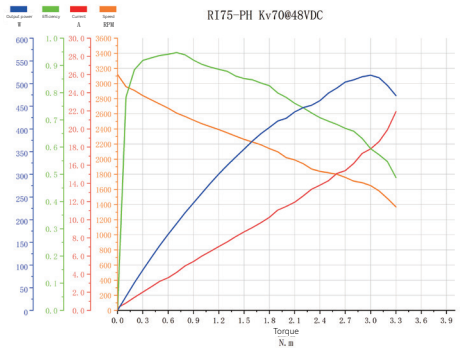
RI75-PH 48
RI85-PH 49
RI115-PH 50

RI75-PH KV70, Φ70*31.8mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

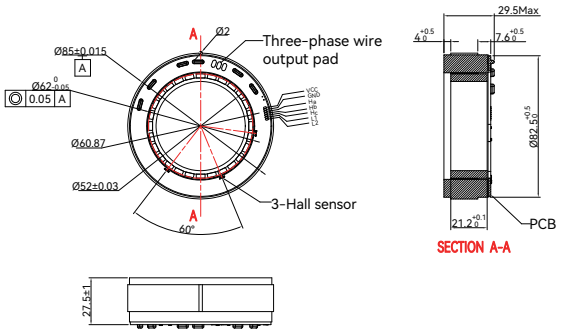
Hall Distribution Angle	120 Electriacl Degrees
Dielectric Strength	1000VAC/1S
Insulation Resistance	100MΩ，500VDC
Operating Environment	-20℃~+85℃
Insulation Class	Class H

ELECTRIC PARAMETERS

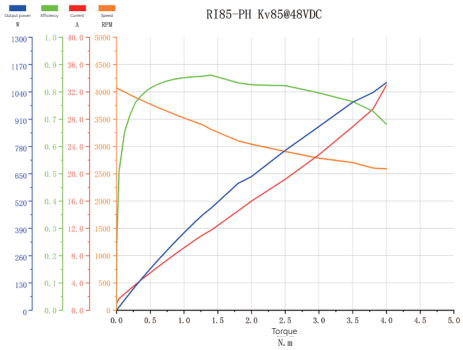
Motor Weight (g)	348
Voltage (V)	48
Continuous Torque (N·m)	1.2
Peak Torque (N·m)	3.8
Max Speed at Continuous Torque (RPM)	2365
Continuous Current (A)	7.28
Peak Current (A)	24.2 (Ref)
Torque Constant (N·m/A)	0.143
Back EMF Constant (V/RPM)	0.015
Speed Constant (RPM/V)	70
Motor Constant (N·m/√w)	0.203
Inertia (g·cm²)	416
Number of Pole Pairs	14
Phase Resistance (mΩ)	495
Phase Inductance (μH)	673
Maximum Torque-to-Weight Ratio (N·m/kg)	11.8
Electrical Time Constant (ms)	0.0013
Electromechanical Time Constant (ms)	0.001

RI85-PH KV85, Φ85*27.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

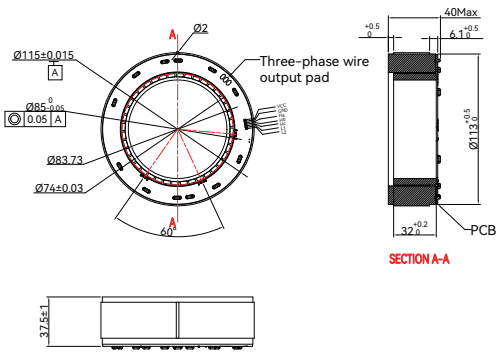
Hall Distribution Angle	120 Electriacl Degrees
Dielectric Strength	1000VAC/1S
Insulation Resistance	100MΩ，500VDC
Operating Environment	-20℃~+85℃
Insulation Class	Class H

ELECTRIC PARAMETERS

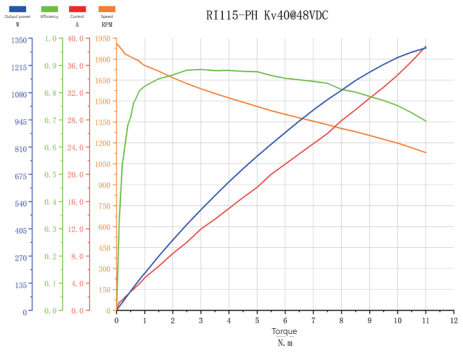
Motor Weight (g)	411
Voltage (V)	48
Continuous Torque (N·m)	2
Peak Torque (N·m)	5
Max Speed at Continuous Torque (RPM)	3042
Continuous Current (A)	16
Peak Current (A)	44 (Ref)
Torque Constant (N·m/A)	0.13
Back EMF Constant (V/RPM)	0.011
Speed Constant (RPM/V)	85
Motor Constant (N·m/√w)	0.299
Inertia (g·cm²)	652
Number of Pole Pairs	14
Phase Resistance (mΩ)	138.5
Phase Inductance (μH)	239.4
Maximum Torque-to-Weight Ratio (N·m/kg)	11.6
Electrical Time Constant (ms)	0.0017
Electromechanical Time Constant (ms)	0.0007

RI115-PH KV40, Φ115*38.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

Hall Distribution Angle	120 Electriacl Degrees
Dielectric Strength	1000VAC/1S
Insulation Resistance	100MΩ，500VDC
Operating Environment	-20℃~+85℃
Insulation Class	Class H

ELECTRIC PARAMETERS

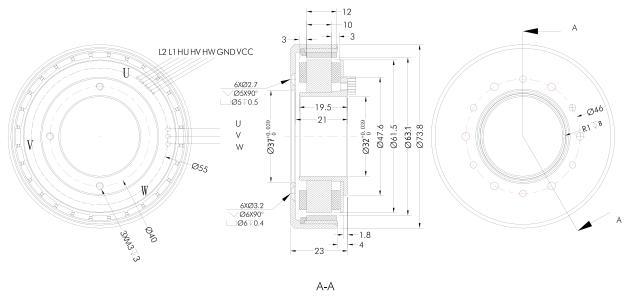
Motor Weight (g)	1108
Voltage (V)	48
Continuous Torque (N·m)	5.5
Peak Torque (N·m)	16
Max Speed at Continuous Torque (RPM)	1430
Continuous Current (A)	17.8
Peak Current (A)	52 (Ref)
Torque Constant (N·m/A)	0.29
Back EMF Constant (V/RPM)	0.025
Speed Constant (RPM/V)	40
Motor Constant (N·m/√w)	0.66
Inertia (g·cm²)	3461
Number of Pole Pairs	20
Phase Resistance (mΩ)	134
Phase Inductance (μH)	245
Maximum Torque-to-Weight Ratio (N·m/kg)	15
Electrical Time Constant (ms)	0.0018
Electromechanical Time Constant (ms)	0.0008

RO Series Frameless Outrunner Torque Motor

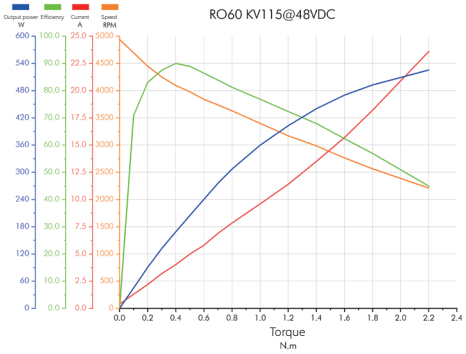
RO60 52
RO80 53
RO100 54

RO60 KV115, Φ73.8*23mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

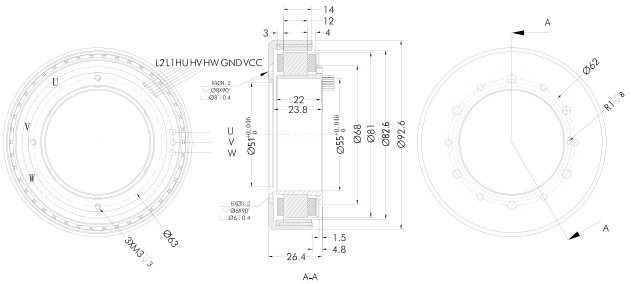
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

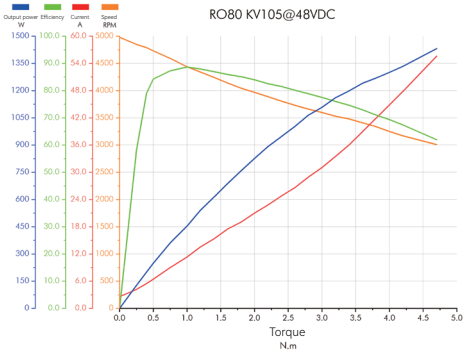
Rated Voltage (V)	48
No-Load Speed (RPM)	5230
Rated Torque (N·m)	0.8
Rated Speed (RPM)	4200
Rated Current (A)	8.5
Peak Torque (N·m)	2.4
Peak Current (A)	40
Kv (RPM/V)	115
Kt (N·m/A)	0.094
Ke (V/kRPM)	8.28
Phase to Phase Resistance (mΩ)	300
Phase to Phase Inductance (μH)	395
Inertia (g·cm²)	840
Km (N·m/√W)	0.17
Mechanical Time Constant (ms)	2.86
Electrical Time Constant (ms)	1.32
Weight (g)	248
Maximum Torque Weight Ratio (N·m/kg)	10

RO80 KV105, Φ92.6*26.4mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

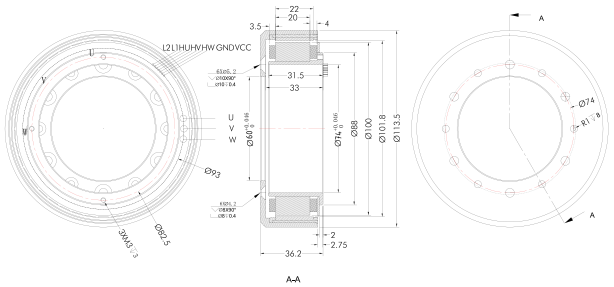
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Class	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

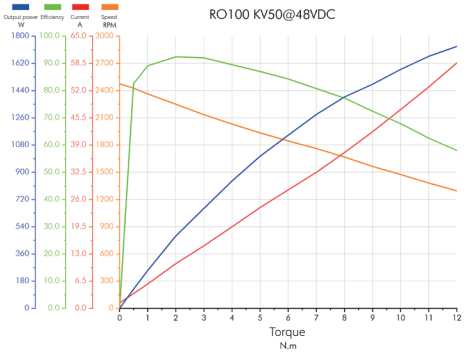
Rated Voltage (V)	48
No-Load Speed (RPM)	4800
Rated Torque (N·m)	1.3
Rated Speed (RPM)	4000
Rated Current (A)	15
Peak Torque (N·m)	4.5
Peak Current (A)	55
Kv (RPM/V)	105
Kt (N·m/A)	0.087
Ke (V/kRPM)	9.07
Phase to Phase Resistance (mΩ)	120
Phase to Phase Inductance (μH)	103
Inertia (g·cm²)	1859
Km (N·m/√W)	0.25
Mechanical Time Constant (ms)	2.95
Electrical Time Constant (ms)	0.86
Weight (g)	345
Maximum Torque Weight Ratio (N·m/kg)	13

RO100 KV50, Φ113.5*36.2mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

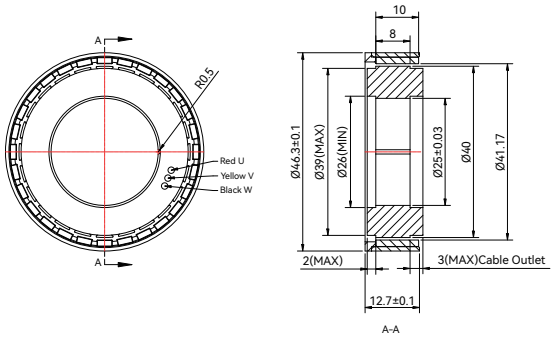
Rated Voltage (V)	48
No-Load Speed (RPM)	2450
Rated Torque (N·m)	4
Rated Speed (RPM)	2000
Rated Current (A)	20
Peak Torque (N·m)	12
Peak Current (A)	62
Kv (RPM/V)	50
Kt (N·m/A)	0.200
Ke (V/kRPM)	18.67
Phase to Phase Resistance (mΩ)	143
Phase to Phase Inductance (μH)	137
Inertia (g·cm²)	4845
Km (N·m/√W)	0.53
Mechanical Time Constant (ms)	1.73
Electrical Time Constant (ms)	0.96
Weight (g)	685
Maximum Torque Weight Ratio (N·m/kg)	17

RO LITE Series Frameless Outrunner Torque Motor

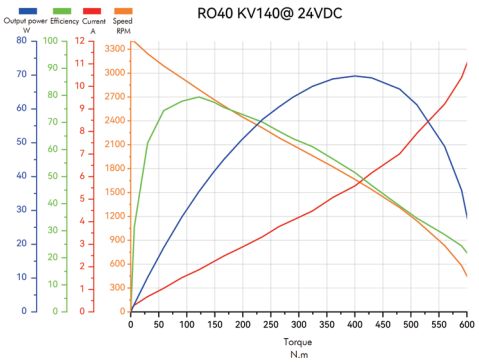
RO40 LITE 56
RO50 LITE 57
RO60 LITE 58
RO80 LITE 59
RO100 LITE 60

RO40 LITE KV140, Φ46.3*13mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

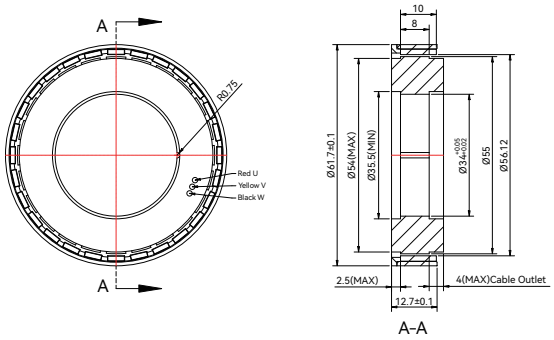
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	-
Insulation Class	F
Insulation High-Voltage (V)	500
Insulation Resistance (mΩ)	10
Cogging Torque (N·m)	6
Pole Pairs	14
Rotor Weight (g)	-
Lead-Out	-

ELECTRIC PARAMETERS

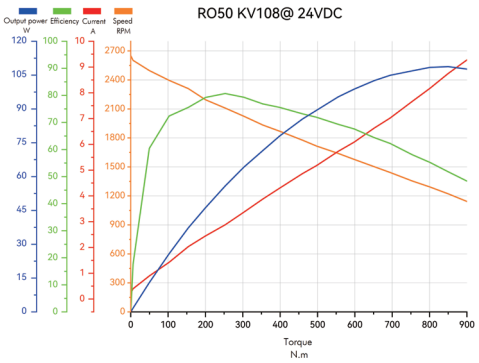
Rated Voltage (V)	24
No-Load Speed (RPM)	3400
Rated Torque (N·m)	0.2
Rated Speed (RPM)	2450
Rated Current (A)	2.85
Peak Torque (N·m)	0.6
Peak Current (A)	11.6
Kv (RPM/V)	141
Kt (N·m/A)	0.068
Stator Weight (g)	-
Ke (V/kRPM)	7
Phase to Phase Resistance (mΩ)	1700
Phase to Phase Inductance (μH)	580
Inertia (g·cm²)	122.11
Km (N·m/√W)	0.52
Mechanical Time Constant (ms)	2.2
Electrical Time Constant (ms)	0.34
Weight (g)	60

RO50 LITE KV108, Φ61.7*14.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

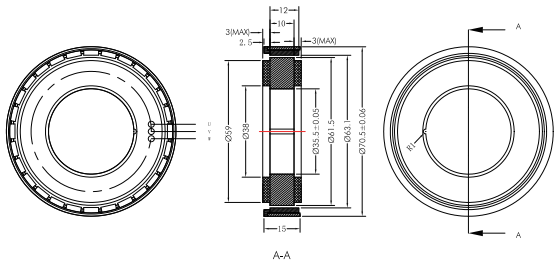
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	-
Insulation Class	F
Insulation High-Voltage (V)	500
Insulation Resistance (mΩ)	10
Cogging Torque (N·m)	10
Pole Pairs	14
Rotor Weight (g)	-
Lead-Out	-

ELECTRIC PARAMETERS

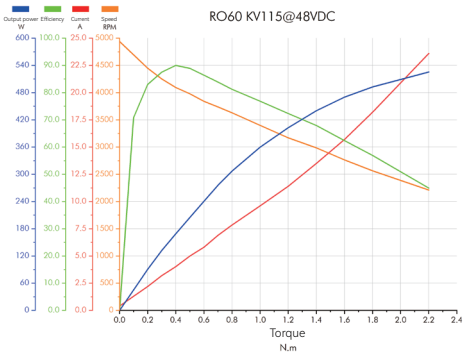
Rated Voltage (V)	24
No-Load Speed (RPM)	2600
Rated Torque (N·m)	0.3
Rated Speed (RPM)	2100
Rated Current (A)	3.4
Peak Torque (N·m)	0.9
Peak Current (A)	0.9
Kv (RPM/V)	108
Kt (N·m/A)	0.088
Stator Weight (g)	-
Ke (V/kRPM)	9.2
Phase to Phase Resistance (mΩ)	1035
Phase to Phase Inductance (μH)	460
Inertia (g·cm²)	327.21
Km (N·m/√W)	0.087
Mechanical Time Constant (ms)	2.4
Electrical Time Constant (ms)	0.44
Weight (g)	105

RO60 LITE KV115, Φ70.5*15mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

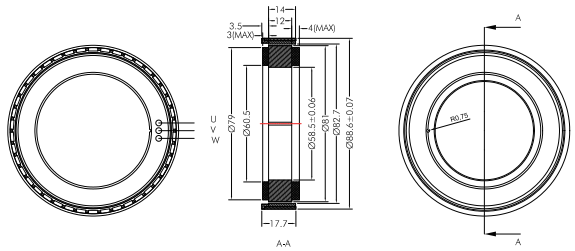
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	F
Insulation High-Voltage (V)	500
Insulation Resistance (mΩ)	10
Cogging Torque (N·m)	15
Pole Pairs	14
Rotor Weight (g)	70
Lead-Out	Three-Phase Motor Lead-Out: Enameled Wire Straight Out 100±5mm Tinned 5±2mm

ELECTRIC PARAMETERS

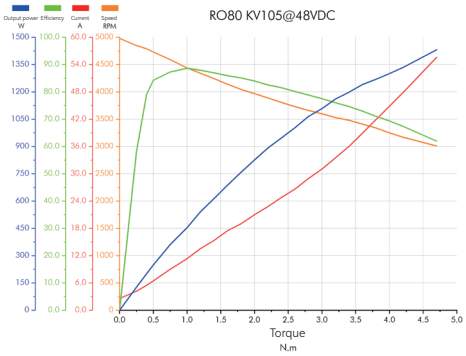
Rated Voltage (V)	48
No-Load Speed (RPM)	5520
Rated Torque (N·m)	0.8
Rated Speed (RPM)	4200
Rated Current (A)	8.5
Peak Torque (N·m)	2.4
Peak Current (A)	40
Kv (RPM/V)	115
Kt (N·m/A)	0.094
Stator Weight (g)	125
Ke (V/kRPM)	8.28
Phase to Phase Resistance (mΩ)	300
Phase to Phase Inductance (μH)	395
Inertia (g·cm²)	800
Km (N·m/√W)	0.17
Mechanical Time Constant (ms)	2.86
Electrical Time Constant (ms)	1.32
Weight (g)	195

RO80 LITE KV105, Φ88.6*17.7mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

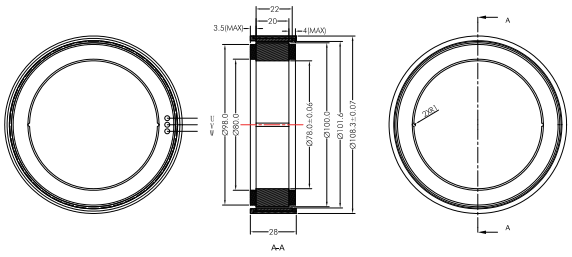
Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20~50
Winding Type	delta
Insulation Class	F
Insulation High-Voltage (V)	500
Insulation Resistance (mΩ)	10
Cogging Torque (N·m)	24
Pole Pairs	21
Rotor Weight (g)	85
Lead-Out	Three-Phase Motor Lead-Out: Enameled Wire Straight Out 100±5mm Tinned 5±2mm

ELECTRIC PARAMETERS

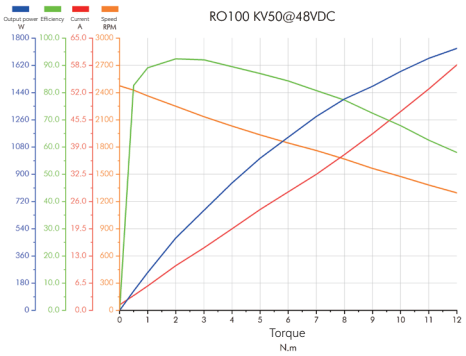
Rated Voltage (V)	48
No-Load Speed (RPM)	5040
Rated Torque (N·m)	1.3
Rated Speed (RPM)	3600
Rated Current (A)	15
Peak Torque (N·m)	4
Peak Current (A)	50
Kv (RPM/V)	105
Kt (N·m/A)	0.087
Stator Weight (g)	180
Ke (V/kRPM)	9.07
Phase to Phase Resistance (mΩ)	120
Phase to Phase Inductance (μH)	103
Inertia (g·cm²)	1600
Km (N·m/√W)	0.25
Mechanical Time Constant (ms)	2.95
Electrical Time Constant (ms)	0.86
Weight (g)	265

RO100 LITE KV55, Φ108.3*28mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



CHARACTERISTICS OVERVIEW

Application	Cobot Arm, Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20~50
Winding Type	star
Insulation Class	F
Insulation High-Voltage (V)	500
Insulation Resistance (mΩ)	10
Cogging Torque (N·m)	55
Pole Pairs	21
Rotor Weight (g)	190
Lead-Out	Three-Phase Motor Lead-Out: Enameled Wire Straight Out 100±5mm Tinned 5±2mm

ELECTRIC PARAMETERS

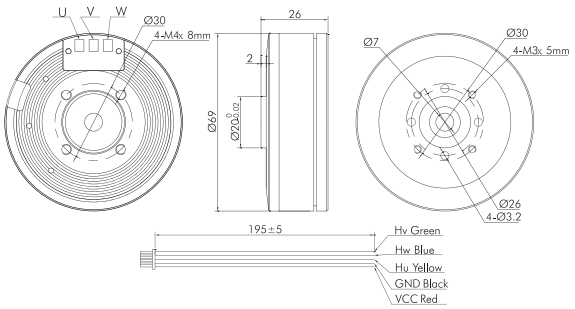
Rated Voltage (V)	48
No-Load Speed (RPM)	2550
Rated Torque (N·m)	4
Rated Speed (RPM)	2000
Rated Current (A)	20
Peak Torque (N·m)	12
Peak Current (A)	62
Kv (RPM/V)	53
Kt (N·m/A)	0.2
Stator Weight (g)	335
Ke (V/kRPM)	18.67
Phase to Phase Resistance (mΩ)	143
Phase to Phase Inductance (μH)	137
Inertia (g·cm²)	5290
Km (N·m/√W)	0.53
Mechanical Time Constant (ms)	1.73
Electrical Time Constant (ms)	0.96
Weight (g)	525

R Series Outrunner Torque Motor

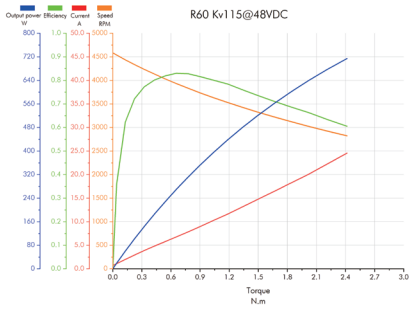
R60 62
R80 63
R100 64

R60 KV115, Φ69*26mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	F
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

CONNECTOR

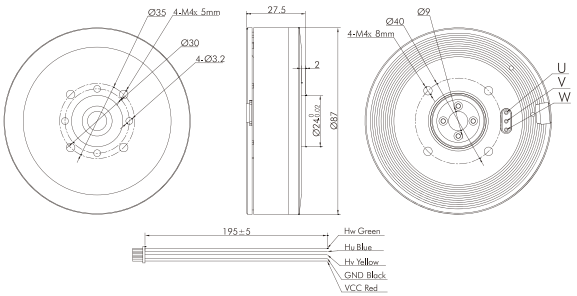
Motor Phase U V W	Reference Engineering Drawing
Hall Sensor Port VCC, GND, Hu, Hv, Hw	Reference Engineering Drawing
Correspondence	Hu-U Hv-V Hw-W

ELECTRIC PARAMETERS

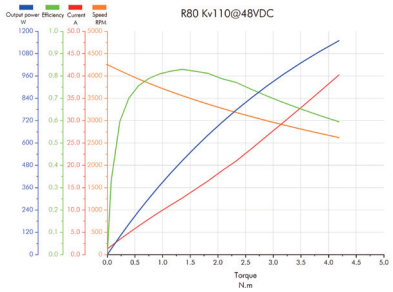
Rated Voltage (V)	48	Ke (V/kRPM)	9.82
No-Load Speed (RPM)	4656	Phase to Phase Resistance (mΩ)	235
Rated Torque (N·m)	0.75	Phase to Phase Inductance (μH)	183
Rated Speed (RPM)	3845	Inertia (g·cm²)	363.74
Rated Current (A)	7.6	Km (N·m/√W)	0.2022
Peak Torque (N·m)	2.3	Mechanical Time Constant (ms)	0.89
Peak Current (A)	23.2	Electrical Time Constant (ms)	0.78
Kv (RPM/V)	97	Weight (g)	248
Kt (N·m/A)	0.098	Maximum Torque Weight Ratio (N·m/kg)	9.3

R80 KV110, Φ87*27.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Class	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

CONNECTOR

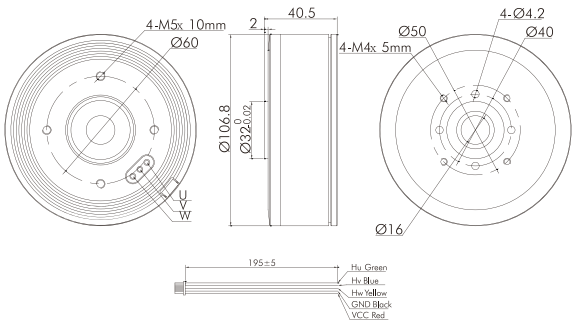
Motor Phase U V W	Reference Engineering Drawing
Hall Sensor Port VCC, GND, Hu, Hv, Hw	Reference Engineering Drawing
Correspondence	Hu-U Hv-V Hw-W

ELECTRIC PARAMETERS

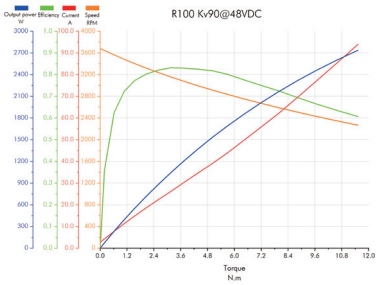
Rated Voltage (V)	48	Ke (V/kRPM)	10.58
No-Load Speed (RPM)	4320	Phase to Phase Resistance (mΩ)	125
Rated Torque (N·m)	1.3	Phase to Phase Inductance (μH)	87
Rated Speed (RPM)	3580	Inertia (g·cm²)	759.98
Rated Current (A)	12.3	Km (N·m/√W)	0.3026
Peak Torque (N·m)	4	Mechanical Time Constant (ms)	0.83
Peak Current (A)	38.6	Electrical Time Constant (ms)	0.70
Kv (RPM/V)	90	Weight (g)	354
Kt (N·m/A)	0.107	Maximum Torque Weight Ratio (N·m/kg)	11.3

R100 KV90, Φ106.8*40.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Exoskeleton
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Delta
Insulation Class	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

CONNECTOR

Motor Phase U V W	Reference Engineering Drawing
Hall Sensor Port VCC, GND, Hu, Hv, Hw	Reference Engineering Drawing
Correspondence	Hu-U Hv-V Hw-W

ELECTRIC PARAMETERS

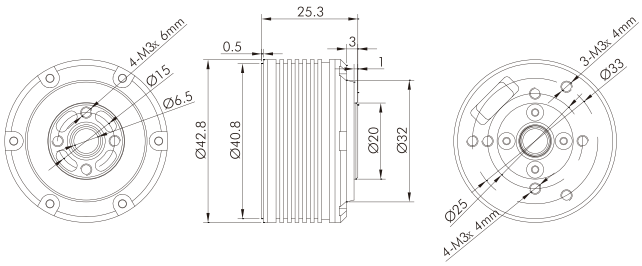
Rated Voltage (V)	48	Ke (V/kRPM)	12.21
No-Load Speed (RPM)	3740	Phase to Phase Resistance (mΩ)	51
Rated Torque (N·m)	3.7	Phase to Phase Inductance (μH)	33
Rated Speed (RPM)	3080	Inertia (g·cm²)	2505.4
Rated Current (A)	30.1	Km (N·m/√W)	0.5447
Peak Torque (N·m)	11	Mechanical Time Constant (ms)	0.84
Peak Current (A)	90.5	Electrical Time Constant (ms)	0.65
Kv (RPM/V)	78	Weight (g)	731
Kt (N·m/A)	0.123	Maximum Torque Weight Ratio (N·m/kg)	16.4

G Series Inrunner Gimbal Motor

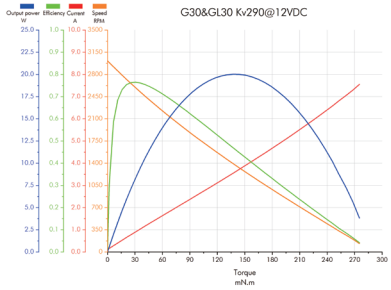
G30 66
G35 67
G40 68
G60 70
G80 72
G100 74

G30 KV290, Φ42.8*25.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

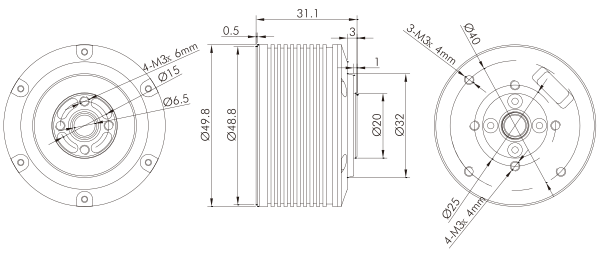
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	7

ELECTRIC PARAMETERS

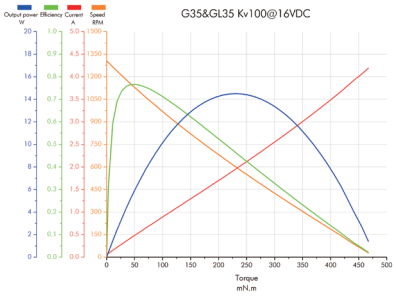
Rated Voltage (V)	12
No-Load Speed (RPM)	3060
Rated Torque (N·m)	0.08
Rated Speed (RPM)	1990
Rated Current (A)	2.13
Peak Torque (N·m)	0.28
Peak Current (A)	7.4
Kv (RPM/V)	255
Kt (N·m/A)	0.038
Ke (V/kRPM)	3.73
Phase to Phase Resistance (mΩ)	1530
Phase to Phase Inductance (μH)	330
Inertia (g·cm^2)	24.2
Km (N·m/√W)	0.0307
Mechanical Time Constant (ms)	2.56
Electrical Time Constant (ms)	0.22
Weight (g)	41
Maximum Torque Weight Ratio (N·m/kg)	6.83

G35 KV100, Φ49.8*31.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

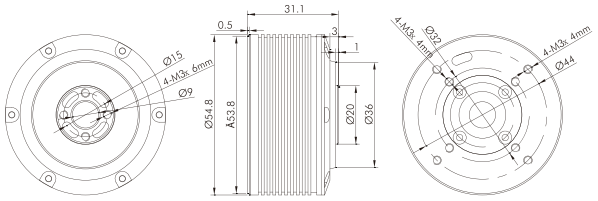
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	7

ELECTRIC PARAMETERS

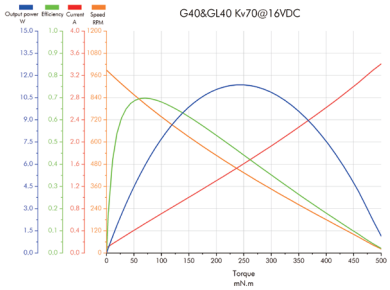
Rated Voltage (V)	16
No-Load Speed (RPM)	1320
Rated Torque (N·m)	0.15
Rated Speed (RPM)	815
Rated Current (A)	1.3
Peak Torque (N·m)	0.46
Peak Current (A)	4
Kv (RPM/V)	82.5
Kt (N·m/A)	0.115
Ke (V/kRPM)	11.54
Phase to Phase Resistance (mΩ)	3600
Phase to Phase Inductance (μH)	2100
Inertia (g·cm^2)	61
Km (N·m/√W)	0.0606
Mechanical Time Constant (ms)	1.66
Electrical Time Constant (ms)	0.58
Weight (g)	90
Maximum Torque Weight Ratio (N·m/kg)	5.11

G40 KV70, Φ54.8*31.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

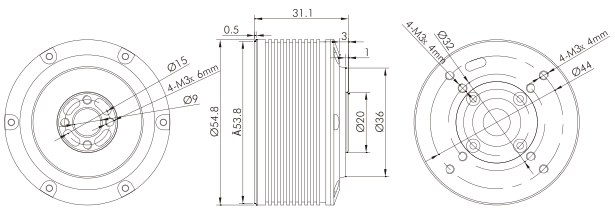
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

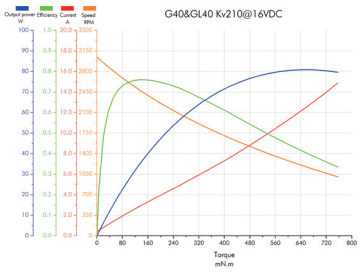
Rated Voltage (V)	16
No-Load Speed (RPM)	1015
Rated Torque (N·m)	0.25
Rated Speed (RPM)	430
Rated Current (A)	1.62
Peak Torque (N·m)	0.5
Peak Current (A)	3.3
Kv (RPM/V)	70
Kt (N·m/A)	0.150
Ke (V/kRPM)	15.00
Phase to Phase Resistance (mΩ)	4500
Phase to Phase Inductance (μH)	1800
Inertia (g·cm^2)	74
Km (N·m/√W)	0.0707
Mechanical Time Constant (ms)	1.48
Electrical Time Constant (ms)	0.40
Weight (g)	107
Maximum Torque Weight Ratio (N·m/kg)	4.67

G40 KV210, Φ54.8*31.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

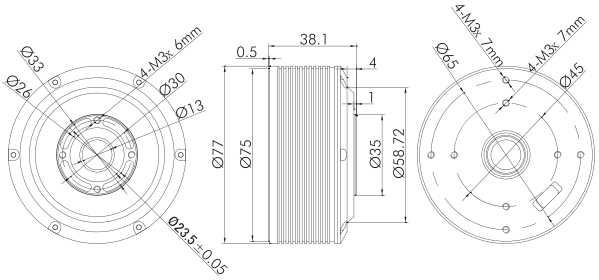
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

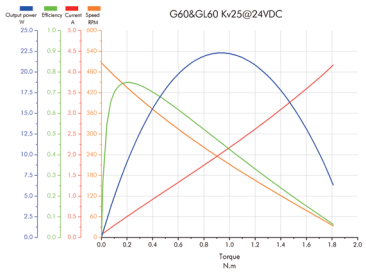
Rated Voltage (V)	16
No-Load Speed (RPM)	3120
Rated Torque (N·m)	0.25
Rated Speed (RPM)	2100
Rated Current (A)	4.9
Peak Torque (N·m)	0.75
Peak Current (A)	14.9
Kv (RPM/V)	210
Kt (N·m/A)	0.050
Ke (V/kRPM)	4.88
Phase to Phase Resistance (mΩ)	500
Phase to Phase Inductance (μH)	180
Inertia (g·cm²)	74
Km (N·m/√W)	0.0707
Mechanical Time Constant (ms)	1.48
Electrical Time Constant (ms)	0.36
Weight (g)	107
Maximum Torque Weight Ratio (N·m/kg)	7.01

G60 KV25, Φ77*38.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

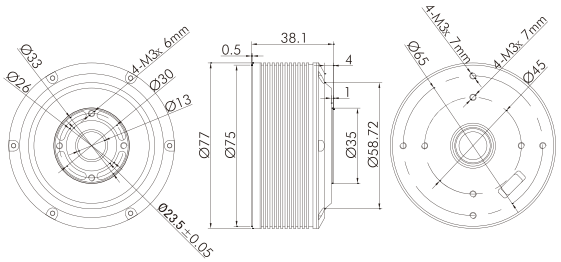
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

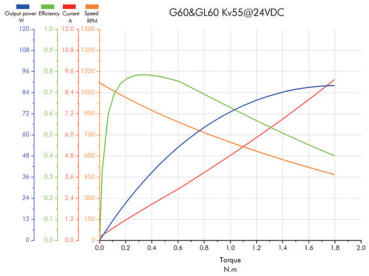
Rated Voltage (V)	24
No-Load Speed (RPM)	516
Rated Torque (N·m)	0.6
Rated Speed (RPM)	310
Rated Current (A)	1.35
Peak Torque (N·m)	1.75
Peak Current (A)	4
Kv (RPM/V)	25
Kt (N·m/A)	0.450
Ke (V/kRPM)	44.30
Phase to Phase Resistance (mΩ)	5500
Phase to Phase Inductance (μH)	2720
Inertia (g·cm²)	355
Km (N·m/√W)	0.1919
Mechanical Time Constant (ms)	0.93
Electrical Time Constant (ms)	0.49
Weight (g)	230
Maximum Torque Weight Ratio (N·m/kg)	7.61

G60 KV55, Φ77*38.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

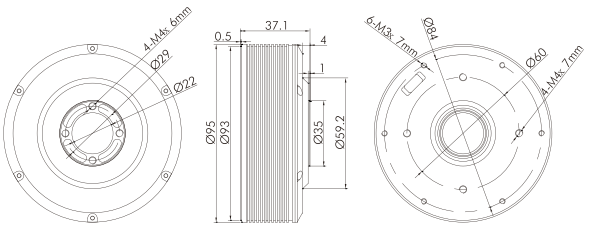
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

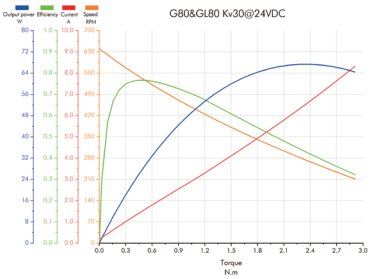
Rated Voltage (V)	24
No-Load Speed (RPM)	1140
Rated Torque (N·m)	0.6
Rated Speed (RPM)	840
Rated Current (A)	2.93
Peak Torque (N·m)	1.75
Peak Current (A)	8.9
Kv (RPM/V)	55
Kt (N·m/A)	0.205
Ke (V/kRPM)	20.05
Phase to Phase Resistance (mΩ)	1200
Phase to Phase Inductance (μH)	900
Inertia (g·cm^2)	355
Km (N·m/√W)	0.1871
Mechanical Time Constant (ms)	1.01
Electrical Time Constant (ms)	0.75
Weight (g)	226
Maximum Torque Weight Ratio (N·m/kg)	7.74

G80 KV30, Φ95*37.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

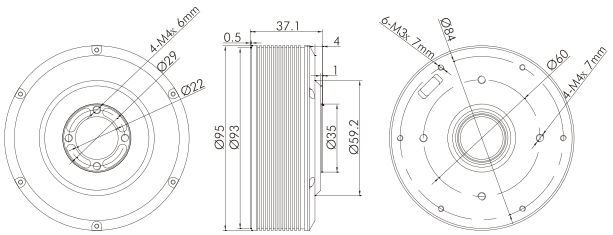
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

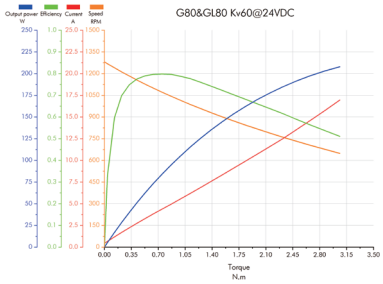
Rated Voltage (V)	16
No-Load Speed (RPM)	650
Rated Torque (N·m)	1
Rated Speed (RPM)	450
Rated Current (A)	2.8
Peak Torque (N·m)	2.9
Peak Current (A)	8.2
Kv (RPM/V)	30
Kt (N·m/A)	0.356
Ke (V/kRPM)	35.27
Phase to Phase Resistance (mΩ)	1800
Phase to Phase Inductance (μH)	1100
Inertia (g·cm^2)	650
Km (N·m/√W)	0.2653
Mechanical Time Constant (ms)	0.92
Electrical Time Constant (ms)	0.61
Weight (g)	315
Maximum Torque Weight Ratio (N·m/kg)	9.21

G80 KV60, Φ95*37.1mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

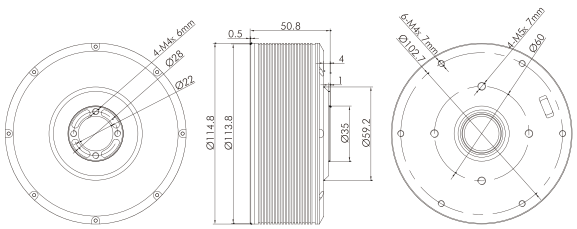
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

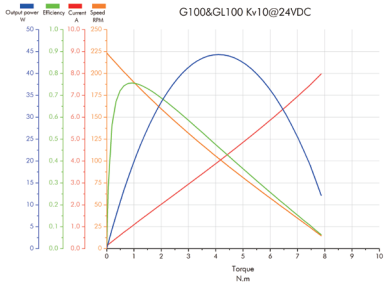
Rated Voltage (V)	24
No-Load Speed (RPM)	1300
Rated Torque (N·m)	1
Rated Speed (RPM)	1010
Rated Current (A)	5.6
Peak Torque (N·m)	2.9
Peak Current (A)	16.3
Kv (RPM/V)	60
Kt (N·m/A)	0.178
Ke (V/kRPM)	17.47
Phase to Phase Resistance (mΩ)	450
Phase to Phase Inductance (μH)	270
Inertia (g·cm^2)	650
Km (N·m/√W)	0.2653
Mechanical Time Constant (ms)	0.92
Electrical Time Constant (ms)	0.60
Weight (g)	315
Maximum Torque Weight Ratio (N·m/kg)	9.21

G100 KV10, Φ114.8*50.8mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Vlass	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	20

ELECTRIC PARAMETERS

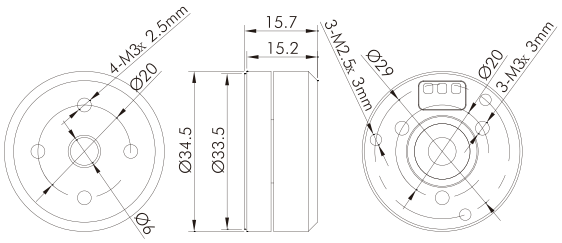
Rated Voltage (V)	224
No-Load Speed (RPM)	223
Rated Torque (N·m)	3
Rated Speed (RPM)	130
Rated Current (A)	2.9
Peak Torque (N·m)	7.7
Peak Current (A)	7.8
Kv (RPM/V)	10
Kt (N·m/A)	1.030
Ke (V/kRPM)	102.4
Phase to Phase Resistance (mΩ)	2650
Phase to Phase Inductance (μH)	2350
Inertia (g·cm^2)	2310
Km (N·m/√W)	0.6327
Mechanical Time Constant (ms)	0.58
Electrical Time Constant (ms)	0.89
Weight (g)	698
Maximum Torque Weight Ratio (N·m/kg)	11.03

GL Series Outrunner Gimbal Motor

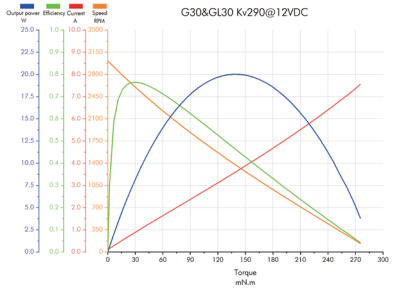
GL30 76
GL35 77
GL40 78
GL40II 80
GL60 81
GL60II 83
GL80 84
GL100 86

GL30 KV290, Φ34.5*15.7mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

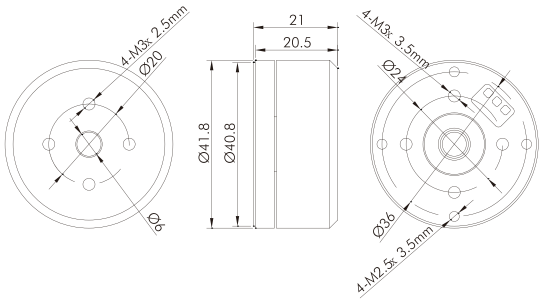
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	7

ELECTRIC PARAMETERS

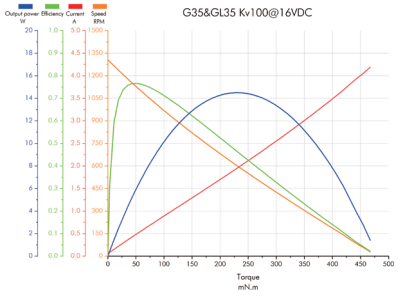
Rated Voltage (V)	12
No-Load Speed (RPM)	3060
Rated Torque (N·m)	0.08
Rated Speed (RPM)	1990
Rated Current (A)	2.13
Peak Torque (N·m)	0.28
Peak Current (A)	7.4
Kv (RPM/V)	290
Kt (N·m/A)	0.038
Ke (V/kRPM)	3.73
Phase to Phase Resistance (mΩ)	1530
Phase to Phase Inductance (μH)	330
Inertia (g·cm^2)	24.2
Km (N·m/√W)	0.0307
Mechanical Time Constant (ms)	2.56
Electrical Time Constant (ms)	0.22
Weight (g)	41
Maximum Torque Weight Ratio (N·m/kg)	6.83

GL35 KV100, Φ41.8*21mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

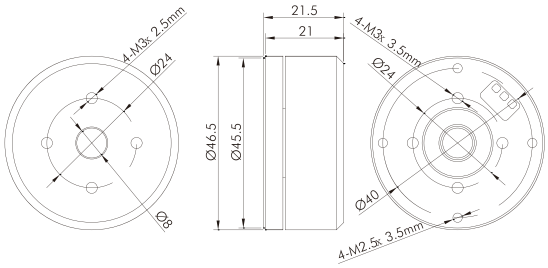
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	7

ELECTRIC PARAMETERS

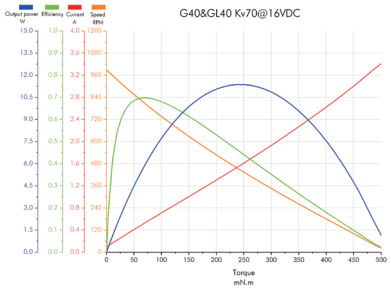
Rated Voltage (V)	16
No-Load Speed (RPM)	1320
Rated Torque (N·m)	0.15
Rated Speed (RPM)	815
Rated Current (A)	1.3
Peak Torque (N·m)	0.46
Peak Current (A)	4
Kv (RPM/V)	100
Kt (N·m/A)	0.115
Ke (V/kRPM)	11.54
Phase to Phase Resistance (mΩ)	3600
Phase to Phase Inductance (μH)	2100
Inertia (g·cm^2)	61
Km (N·m/√W)	0.0606
Mechanical Time Constant (ms)	1.66
Electrical Time Constant (ms)	0.58
Weight (g)	90
Maximum Torque Weight Ratio (N·m/kg)	5.11

GL40 KV70, Φ46.5*21.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

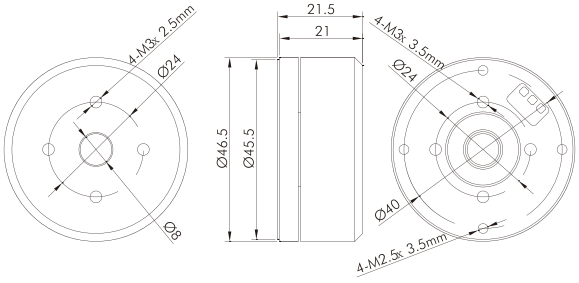
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

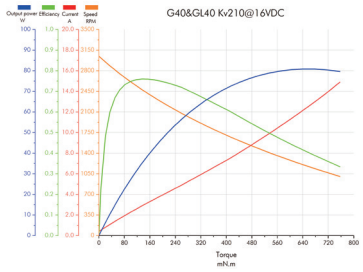
Rated Voltage (V)	16
No-Load Speed (RPM)	1015
Rated Torque (N·m)	0.25
Rated Speed (RPM)	430
Rated Current (A)	1.62
Peak Torque (N·m)	0.5
Peak Current (A)	3.3
Kv (RPM/V)	70
Kt (N·m/A)	0.150
Ke (V/kRPM)	15.00
Phase to Phase Resistance (mΩ)	4500
Phase to Phase Inductance (μH)	1800
Inertia (g·cm^2)	74
Km (N·m/√W)	0.0707
Mechanical Time Constant (ms)	1.48
Electrical Time Constant (ms)	0.40
Weight (g)	107
Maximum Torque Weight Ratio (N·m/kg)	4.67

GL40 KV210, Φ46.5*21.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

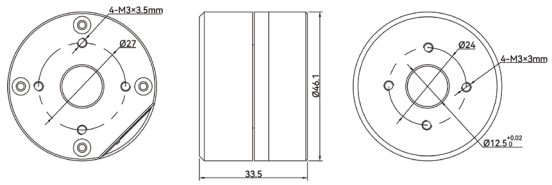
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

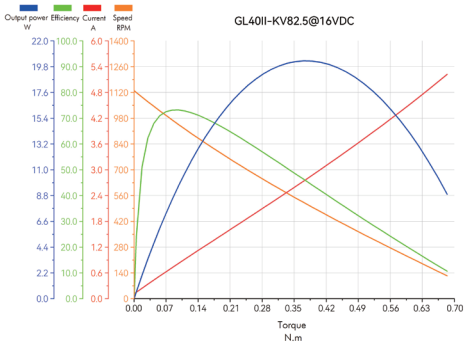
Rated Voltage (V)	16
No-Load Speed (RPM)	3120
Rated Torque (N·m)	0.25
Rated Speed (RPM)	2100
Rated Current (A)	4.9
Peak Torque (N·m)	0.75
Peak Current (A)	14.9
Kv (RPM/V)	210
Kt (N·m/A)	0.050
Ke (V/kRPM)	4.88
Phase to Phase Resistance (mΩ)	500
Phase to Phase Inductance (μH)	180
Inertia (g·cm^2)	74
Km (N·m/√W)	0.0707
Mechanical Time Constant (ms)	1.48
Electrical Time Constant (ms)	0.36
Weight (g)	107
Maximum Torque Weight Ratio (N·m/kg)	7.01

GL40II KV82.5, Φ46.1*33.5mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

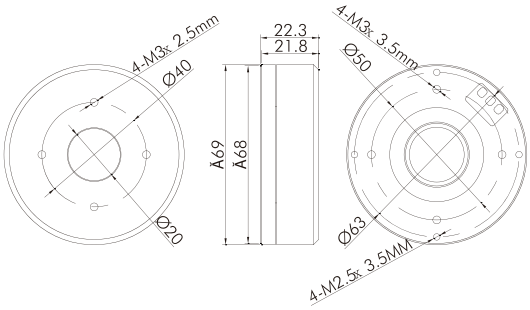
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

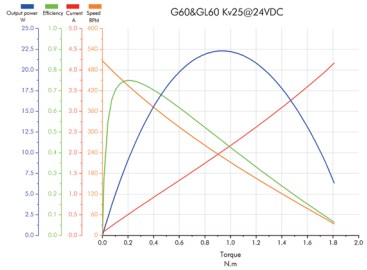
Rated Voltage (V)	16
No-Load Speed (RPM)	1388
Rated Torque (N·m)	0.25
Rated Speed (RPM)	697
Rated Current (A)	1.88
Peak Torque (N·m)	0.68
Peak Current (A)	5.22
Kv (RPM/V)	82.5
Kt (N·m/A)	0.11
Ke (V/kRPM)	0.0115
Phase to Phase Resistance (mΩ)	3000
Phase to Phase Inductance (μH)	1320
Inertia (g·cm^2)	0.794578
Km (N·m/√W)	0.06392
Mechanical Time Constant (ms)	2.045
Electrical Time Constant (ms)	0.44
Weight (g)	112
Maximum Torque Weight Ratio (N·m/kg)	60.71

GL60 KV25, Φ69*22.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

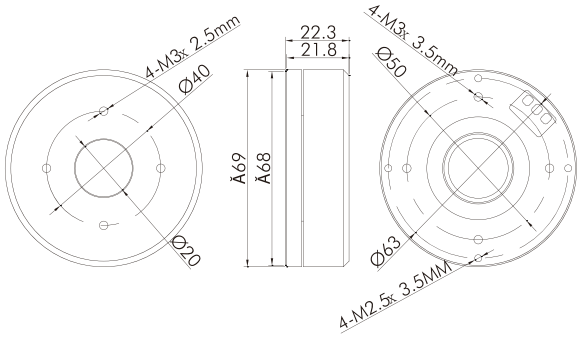
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERS

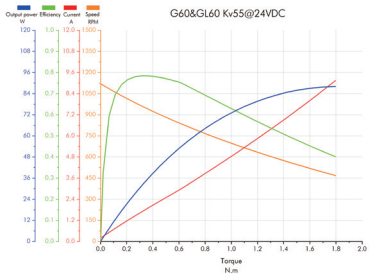
Rated Voltage (V)	24
No-Load Speed (RPM)	516
Rated Torque (N·m)	0.6
Rated Speed (RPM)	310
Rated Current (A)	1.35
Peak Torque (N·m)	1.75
Peak Current (A)	4
Kv (RPM/V)	25
Kt (N·m/A)	0.450
Ke (V/kRPM)	44.30
Phase to Phase Resistance (mΩ)	5500
Phase to Phase Inductance (μH)	2720
Inertia (g·cm^2)	355
Km (N·m/√W)	0.1919
Mechanical Time Constant (ms)	0.96
Electrical Time Constant (ms)	0.49
Weight (g)	230
Maximum Torque Weight Ratio (N·m/kg)	7.61

GL60 KV55, Φ69*22.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

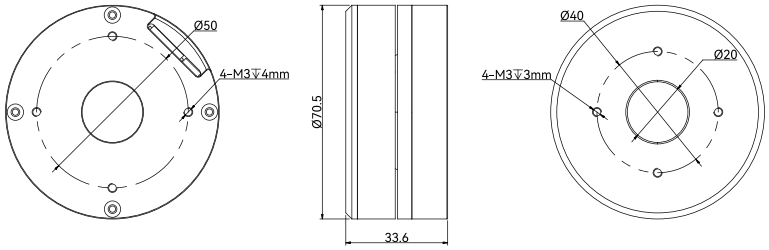
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	500V 5mA/2s
Insulation Resistance	500V 10MΩ
Phase	3
Pole Pairs	14

ELECTRIC PARAMETERSv

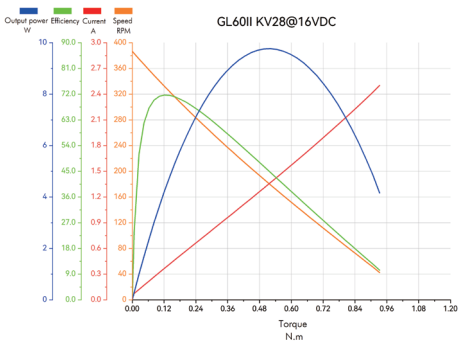
Rated Voltage (V)	24
No-Load Speed (RPM)	1140
Rated Torque (N·m)	0.6
Rated Speed (RPM)	840
Rated Current (A)	2.93
Peak Torque (N·m)	1.75
Peak Current (A)	8.9
Kv (RPM/V)	55
Kt (N·m/A)	0.205
Ke (V/kRPM)	20.05
Phase to Phase Resistance (mΩ)	1200
Phase to Phase Inductance (μH)	900
Inertia (g·cm^2)	355
Km (N·m/√W)	0.1871
Mechanical Time Constant (ms)	1.01
Electrical Time Constant (ms)	0.75
Weight (g)	226
Maximum Torque Weight Ratio (N·m/kg)	7.74

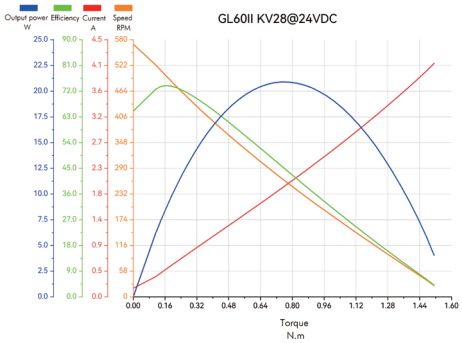
GL60II KV28, Φ70.5*33.6mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION





SPECIFICATIONS

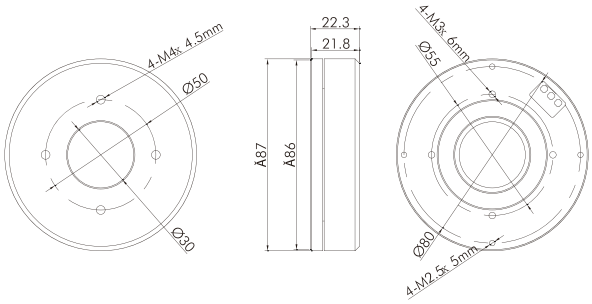
Application	Gimbal, Radar	Bearings	Imported 6705 ZZ
Motor Model	GL60II	Copper Wire	Level E 150℃
Motor Dimensions (mm)	Ø70.5*33.6	Wiring Method	-
Stator Technology	Rust-Proof, 180℃ Coating	Enamelled Wires Temperature Resistance Grade	Level H 180℃
Configuration	24N28P	Coil Withstand Voltage	500V 5mA/2s
Bore Size (mm)	Ø20	Rotor Inertia (g·cm²)	401.086
Torque Constant (N·m/A)	0.34	Waterproof And Dustproof Class	-

Performance Specifications

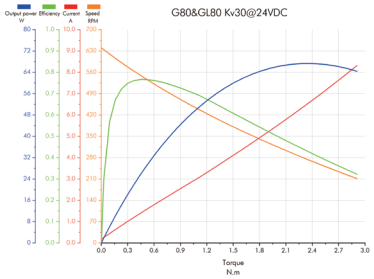
KV	KV28	KV	KV28
Rated Current (A)	1.56	Rated Current (A)	1.54
Rated Torque (N·m)	0.6	Rated Torque (N·m)	0.6
Peak Current (A)	2.75	Peak Current (A)	4.09
Peak Torque (N·m)	1	Peak Torque (N·m)	1.5
Winding Resistance (Ω)	5.8	Winding Resistance (Ω)	5.8
Working Voltage (V)	16	Working Voltage (V)	24
Rated Speed (RPM)	153	Rated Speed (RPM)	319
Line Inductance (mH)	6	Line Inductance (mH)	6
Package Dimensions (mm)	160*122*42	Package Dimensions (mm)	160*122*42
Weight (g)	276	Weight (g)	276

GL80 KV30, Φ87*22.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

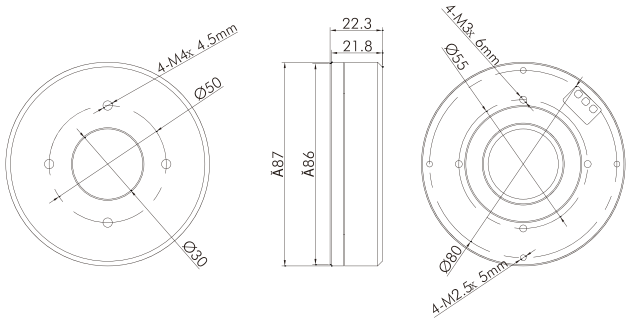
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

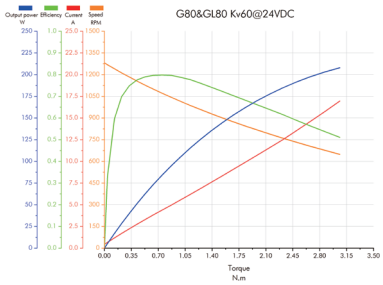
Rated Voltage (V)	24
No-Load Speed (RPM)	650
Rated Torque (N·m)	1
Rated Speed (RPM)	450
Rated Current (A)	2.8
Peak Torque (N·m)	2.9
Peak Current (A)	8.2
Kv (RPM/V)	30
Kt (N·m/A)	0.356
Ke (V/kRPM)	35.27
Phase to Phase Resistance (mΩ)	1800
Phase to Phase Inductance (μH)	1100
Inertia (g·cm^2)	650
Km (N·m/√W)	0.2653
Mechanical Time Constant (ms)	0.92
Electrical Time Constant (ms)	0.61
Weight (g)	315
Maximum Torque Weight Ratio (N·m/kg)	9.21

GL80 KV60, Φ87*22.3mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

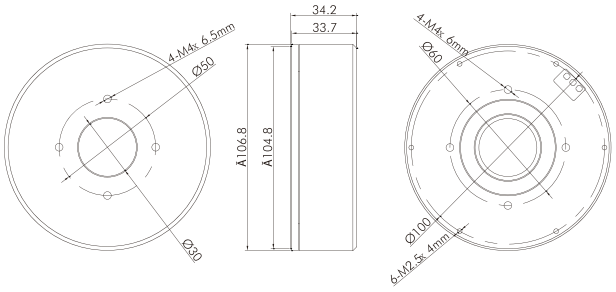
Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	21

ELECTRIC PARAMETERS

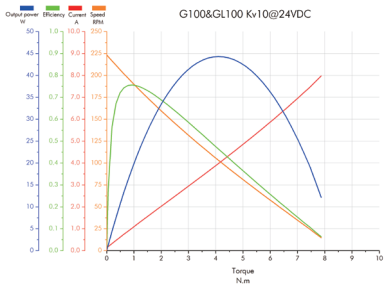
Rated Voltage (V)	24
No-Load Speed (RPM)	1300
Rated Torque (N·m)	1
Rated Speed (RPM)	1010
Rated Current (A)	5.6
Peak Torque (N·m)	2.9
Peak Current (A)	16.3
Kv (RPM/V)	60
Kt (N·m/A)	0.178
Ke (V/kRPM)	17.47
Phase to Phase Resistance (mΩ)	450
Phase to Phase Inductance (μH)	270
Inertia (g·cm^2)	650
Km (N·m/√W)	0.2653
Mechanical Time Constant (ms)	0.92
Electrical Time Constant (ms)	0.60
Weight (g)	315
Maximum Torque Weight Ratio (N·m/kg)	9.21

GL100 KV10, Φ106.8*34.2mm

PRODUCT DRAWING



ANALYTICAL GRAPH OF MOTOR OPERATION



SPECIFICATIONS

Application	Gimbal, Radar
Driving Way	FOC
Operation Ambient Temperature	-20℃~50℃
Winding Type	Star
Insulation Class	H
Insulation High-Voltage	1000V 5mA/2s
Insulation Resistance	1000V 10MΩ
Phase	3
Pole Pairs	20

ELECTRIC PARAMETERS

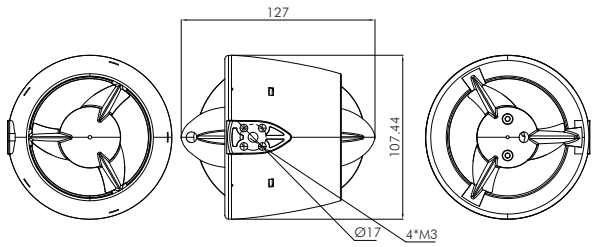
Rated Voltage (V)	24
No-Load Speed (RPM)	223
Rated Torque (N·m)	3
Rated Speed (RPM)	130
Rated Current (A)	2.9
Peak Torque (N·m)	7.7
Peak Current (A)	7.8
Kv (RPM/V)	10
Kt (N·m/A)	1.030
Ke (V/kRPM)	102.4
Phase to Phase Resistance (mΩ)	2650
Phase to Phase Inductance (μH)	2350
Inertia (g·cm^2)	2310
Km (N·m/√W)	0.6327
Mechanical Time Constant (ms)	0.58
Electrical Time Constant (ms)	0.89
Weight (g)	698
Maximum Torque Weight Ratio (N·m/kg)	11.03

Underwater Thruster

W30 88
SW7 90
SW12 91
SW17 92
DW10 93
DW15 94
DW20 95
DW25 96

W30 KV320, Φ127*107.44mm

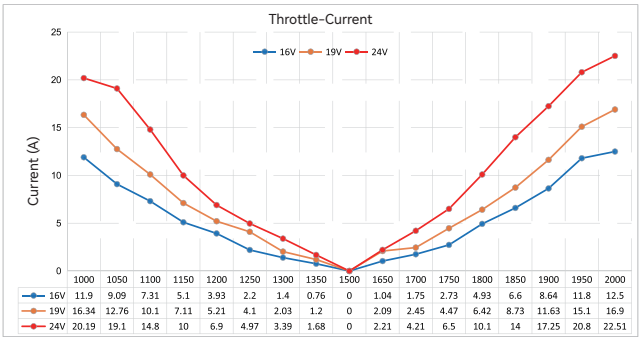
ANALYTICAL GRAPH OF MOTOR OPERATION



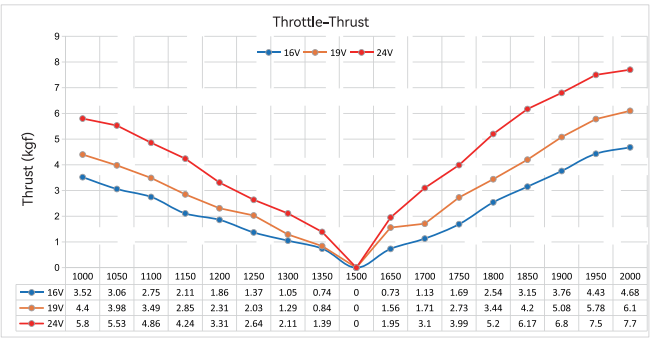
SPECIFICATIONS

Voltage	4~6S
Continuous Current	45A
Instant Current	50A (10s)
Standby Loss	≤50mA
Throttle Flash Frenquency	400Hz
Motor Wires	Black 16AWG 100mm*3P
Power Wires	Red/Black 14AWG 200mm
Signal Wires	PVC JR 3p
Throttle Range	1000~2000us
Weight	420g

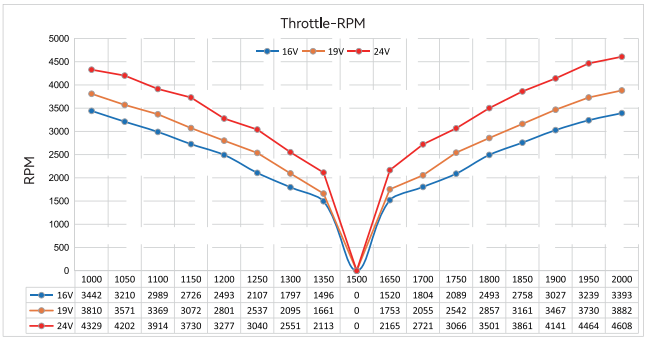
PRODUCT DRAWING



	1000	1050	1100	1150	1200	1250	1300	1350	1500	1650	1700	1750	1800	1850	1900	1950	2000
16V	11.9	9.09	7.31	5.1	3.93	2.2	1.4	0.76	0	1.04	1.75	2.73	4.93	6.6	8.64	11.8	12.5
19V	16.34	12.76	10.1	7.11	5.21	4.1	2.03	1.2	0	2.09	2.45	4.47	6.42	8.73	11.63	15.1	16.9
24V	20.19	19.1	14.8	10	6.9	4.97	3.39	1.68	0	2.21	4.21	6.5	10.1	14	17.25	20.8	22.51



	1000	1050	1100	1150	1200	1250	1300	1350	1500	1650	1700	1750	1800	1850	1900	1950	2000
16V	3.52	3.06	2.75	2.11	1.86	1.37	1.05	0.74	0	0.73	1.13	1.69	2.54	3.15	3.76	4.43	4.68
19V	4.4	3.98	3.49	2.85	2.31	2.03	1.29	0.84	0	1.56	1.71	2.73	3.44	4.2	5.08	5.78	6.1
24V	5.8	5.53	4.86	4.24	3.31	2.64	2.11	1.39	0	1.95	3.1	3.99	5.2	6.17	6.8	7.5	7.7



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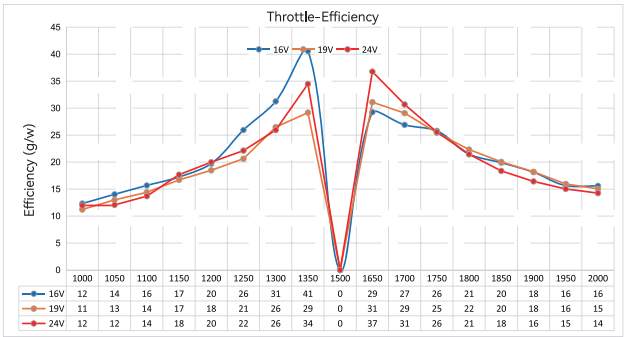
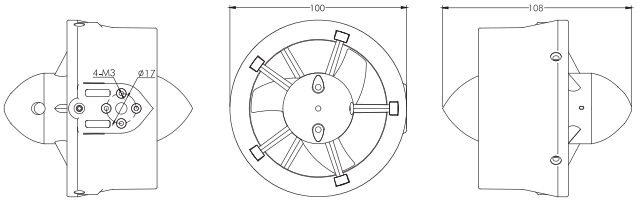
16V	3442	3210	2989	2726	2493	2107	1797	1496	0	1520	1804	2089	2493	2758	3027	3239	3393
19V	3810	3571	3369	3072	2801	2537	2095	1661	0	1753	2055	2542	2857	3161	3467	3730	3882
24V	4329	4202	3914	3730	3277	3040	2551	2113	0	2165	2721	3066	3501	3861	4141	4464	4608



	1000	1050	1100	1150	1200	1250	1300	1350	1500	1650	1700	1750	1800	1850	1900	1950	2000
16V	12	14	16	17	20	26	31	41	0	29	27	26	21	20	18	16	16
19V	11	13	14	17	18	21	26	29	0	31	29	25	22	20	18	16	15
24V	12	12	14	18	20	22	26	34	0	37	31	26	21	18	16	15	14

W30 KV320, Φ100*108mm

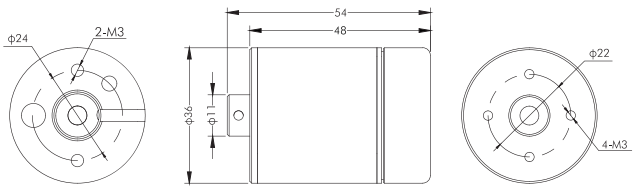
PRODUCT DRAWING



SPECIFICATIONS

Type	W30
Dimension	Ø100*108mm
Forward Thrust	6.7kgf~7.7kgf
Backward Thrust	5.0kgf~5.8kgf
Rate Power	480W
Rated Current	22A
Weight In Air	420g
Weight In Water	200g
Max. Depth	200m
Wire Configuration x Length	18AWG*1000mm
Motor	W30 KV320
Battery (LiPo) Recommend	6S (4.2V)
ESC	W30 ESC-8S-45A
Propeller	W30 Purple PC Propeller
Package Weight	710g

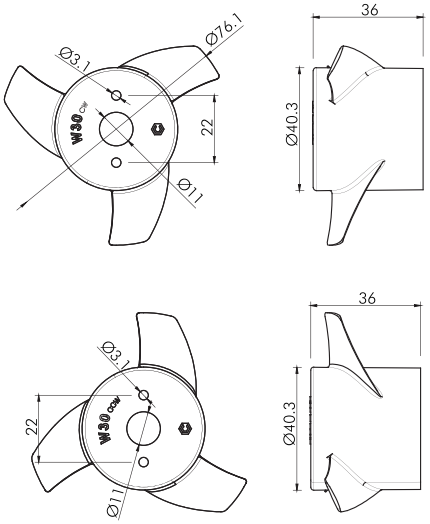
PRODUCT DRAWING



SPECIFICATIONS

W30 Purple PC Propeller	W30 KV320
Rated Current	22A
No-Load Current@10V	0.5A
No-Load RPM@10V	3200RPM
Inductance	125uH
Max. Current	30A/180s
Max. Power	720W/180s
Internal Resistance	220mΩ
Rated Power	480W
Weight	270g
Winding Wire Temperature-Resistant	C级 220℃
Wire Configuration x Length	18AWG*1000mm
Max. Depth	200m
Slot/Pole	12N14P
Winding Hi-pot	500V/5s
Package Weight	580g

PRODUCT DRAWING

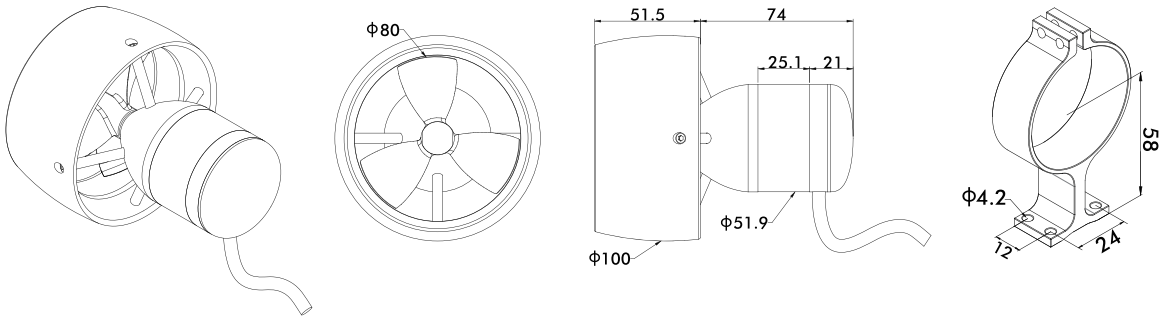


SPECIFICATIONS

Dimension	Ø76.1*36mm
Pitch	3.6英寸
Weight	12g

SW7

PRODUCT DRAWING

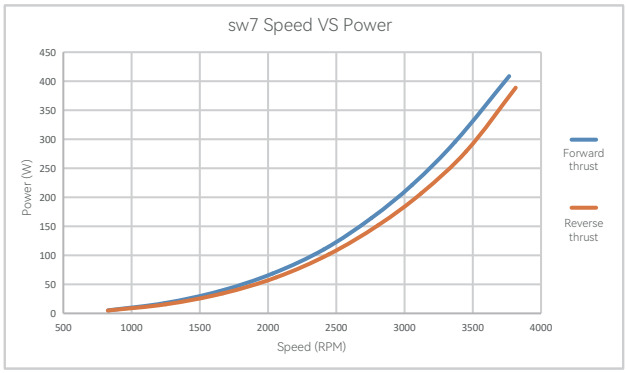


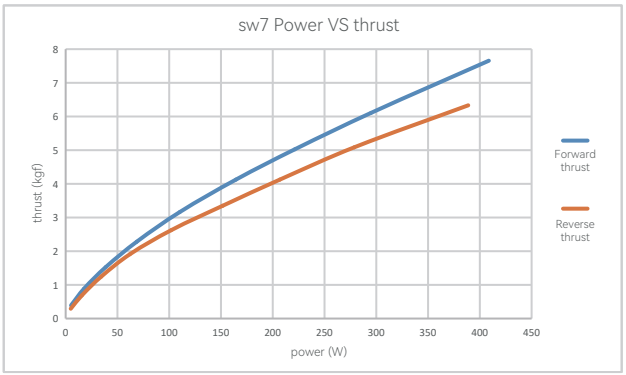
SPECIFICATIONS

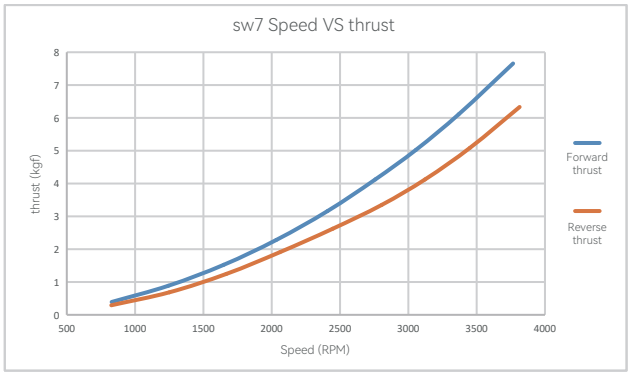
Application	Unmanned Surface Vehicle, Electric Surfboard Handheld Ehruster,Etc
Rated Power	360W
Rated Voltage	36V
Maximum Mooring Thrust	≥7kgf
Applicable Water Depth	0~30M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø80mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Matte) Ø6.5mm-3*18AWG-1.5m
Driver	External
Weight	700g (In Air)/395g (underwater)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION

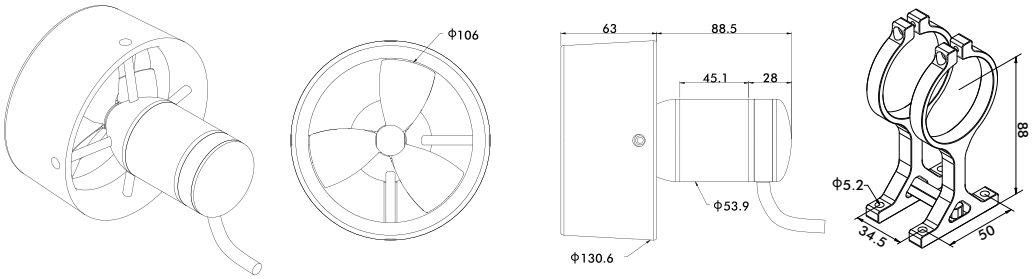






SW12

PRODUCT DRAWING

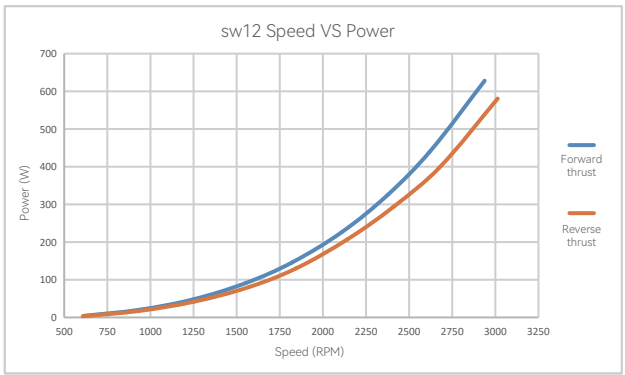


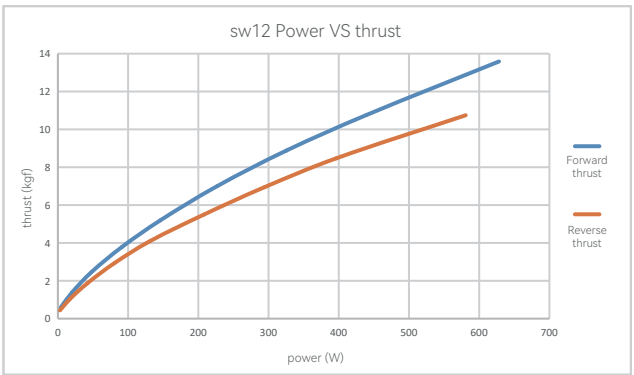
SPECIFICATIONS

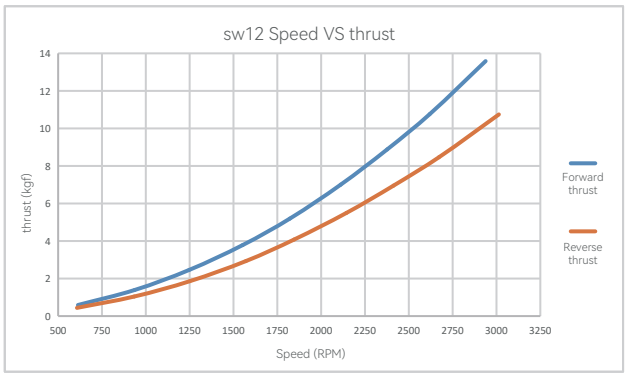
Application	Unmanned Surface Vehicle, Electric Surfboard Handheld Ehruster,Etc
Rated Power	520W
Rated Voltage	48V
Maximum Mooring Thrust	≥12kgf
Applicable Water Depth	0~30M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø106mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Glossy) Ø8mm-3*16AWG-1.5M
Driver	External
Weight	1180g (In Air)/640g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION

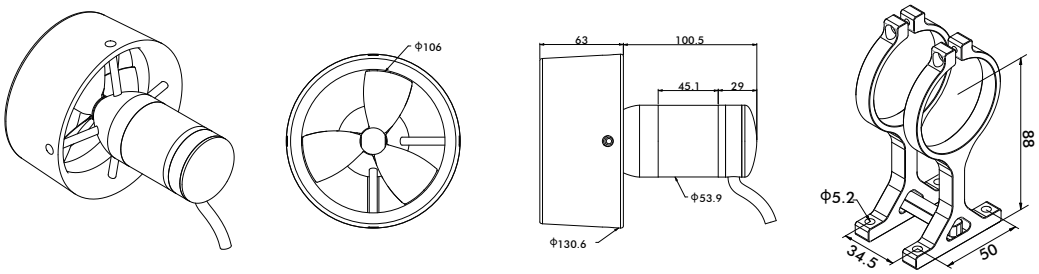






SW17

PRODUCT DRAWING

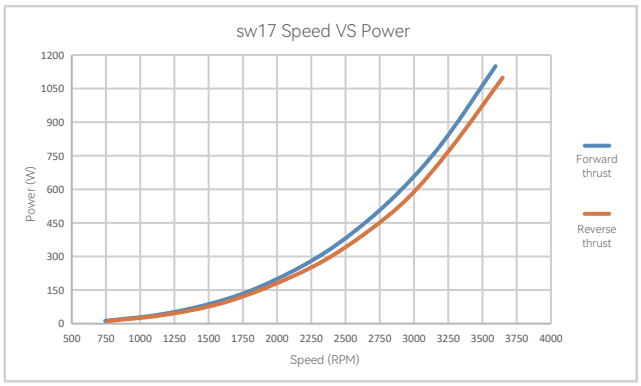


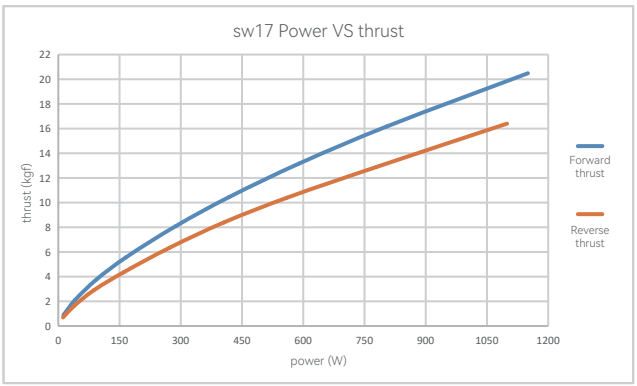
SPECIFICATIONS

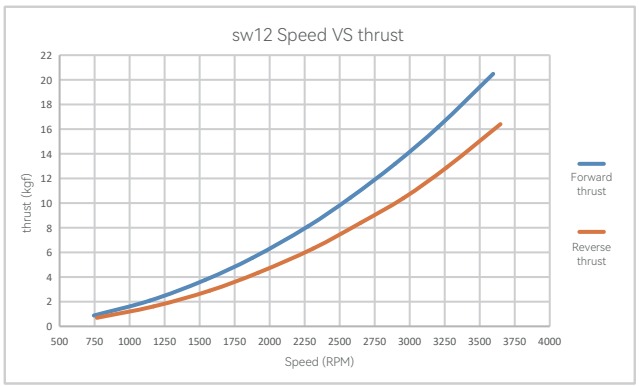
Application	Unmanned Surface Vehicle, Electric Surfboard Handheld Ehruster,Etc
Rated Power	870W
Rated Voltage	48V
Maximum Mooring Thrust	≥17kgf
Applicable Water Depth	0~30M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø106mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Glossy) Ø8.5mm-3*14AWG-1.5M
Driver	External
Weight	1350g (In Air)/775g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION

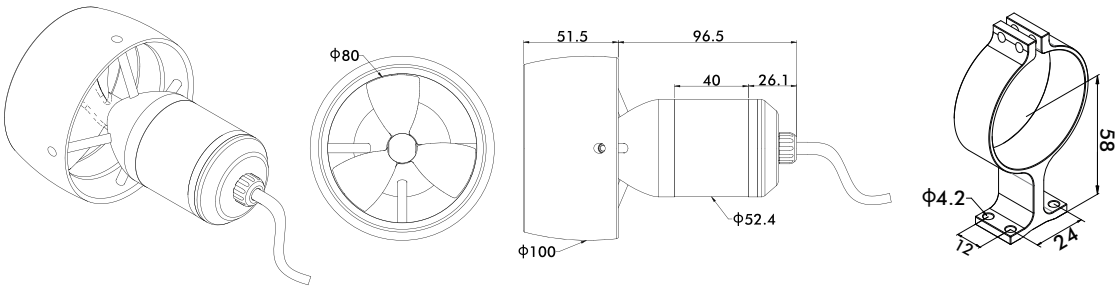






DW10

PRODUCT DRAWING

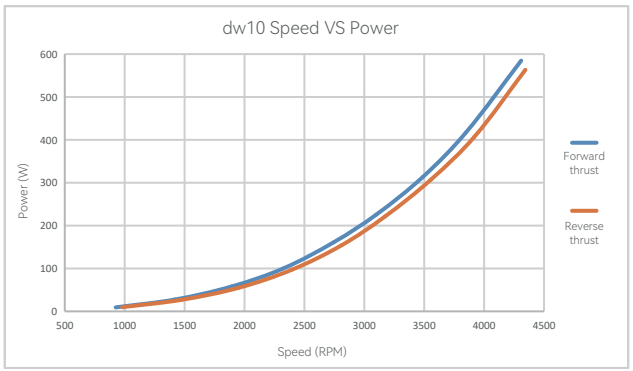


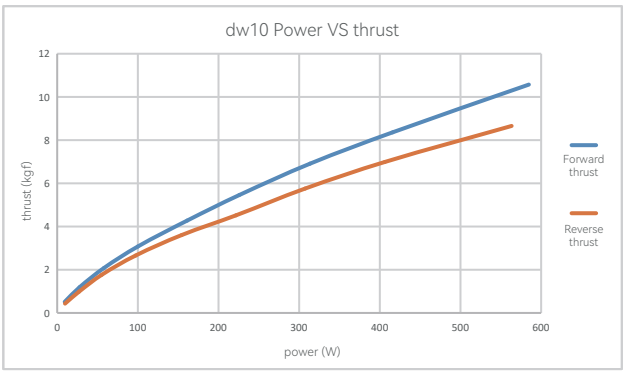
SPECIFICATIONS

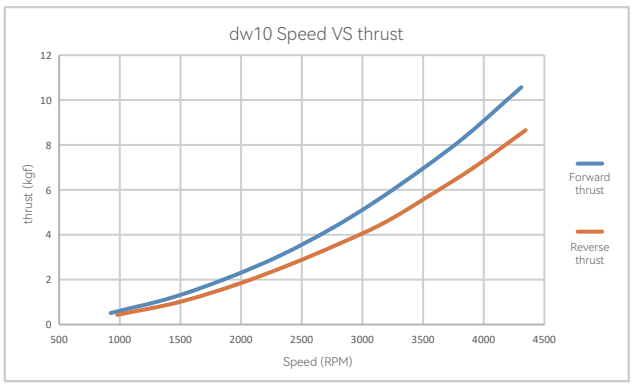
Application	Unmanned Surface Vehicle Electric Surfboard, ROV, Etc
Rated Power	540W
Rated Voltage	36V
Maximum Mooring Thrust	≥10kgf
Applicable Water Depth	0~350M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø80mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Glossy) Ø8mm-3*16AWG-1.5M
Driver	External
Weight	845g (In Air)/480g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION

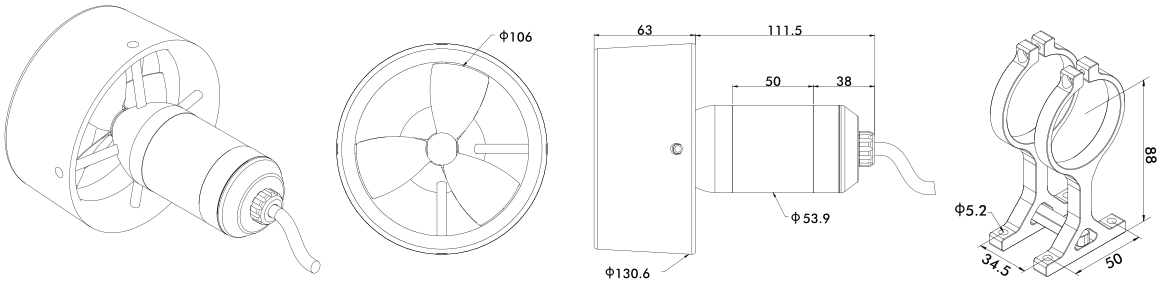






DW15

PRODUCT DRAWING

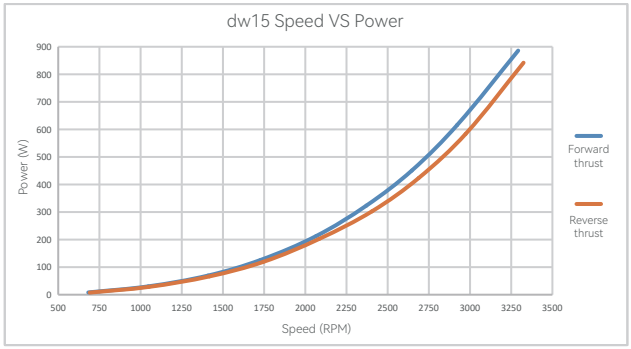


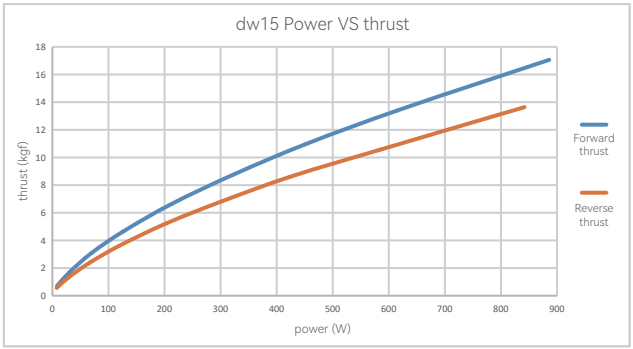
SPECIFICATIONS

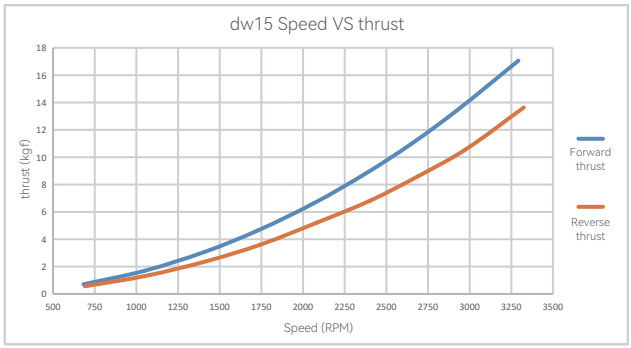
Application	Unmanned Surface Vehicle Electric Surfboard, ROV, Etc
Rated Power	750W
Rated Voltage	48V
Maximum Mooring Thrust	≥15kgf
Applicable Water Depth	0~350M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø106mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Glossy) Ø8mm-3*16AWG-1.5M
Driver	External
Weight	1260g (In Air)/735g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION

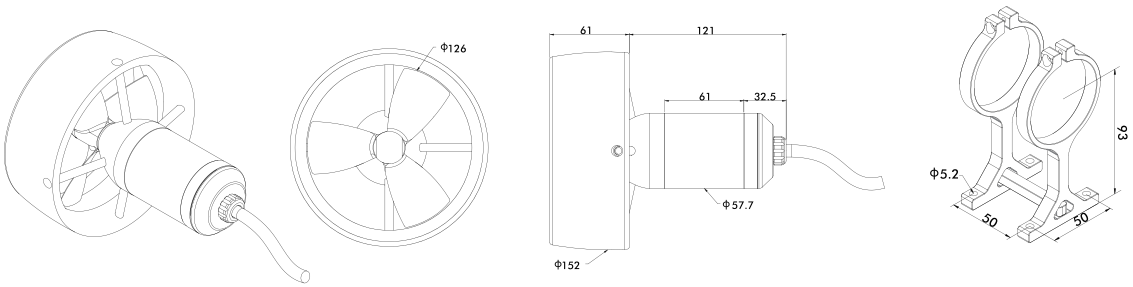






DW20

PRODUCT DRAWING

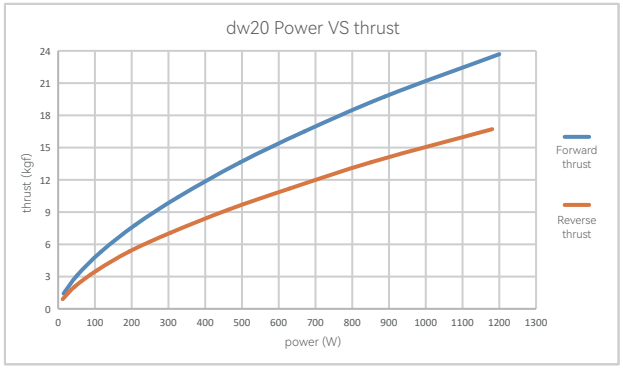


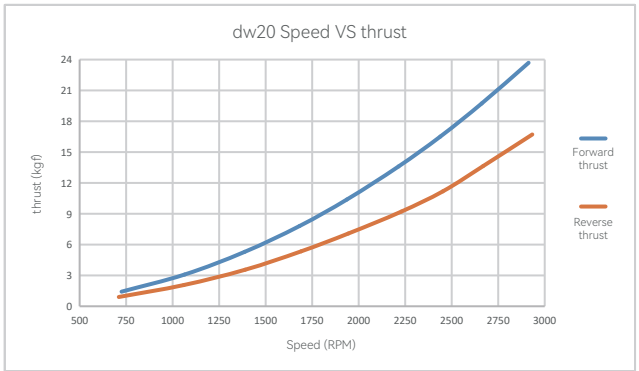
SPECIFICATIONS

Application	Unmanned Surface Vehicle Electric Surfboard, ROV, Etc
Rated Power	910W
Rated Voltage	48V
Maximum Mooring Thrust	≥20kgf
Applicable Water Depth	0~350M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø126mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Matte) Ø9mm-3*13AWG-1.5M
Driver	External
Weight	1630g (In Air)/950g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

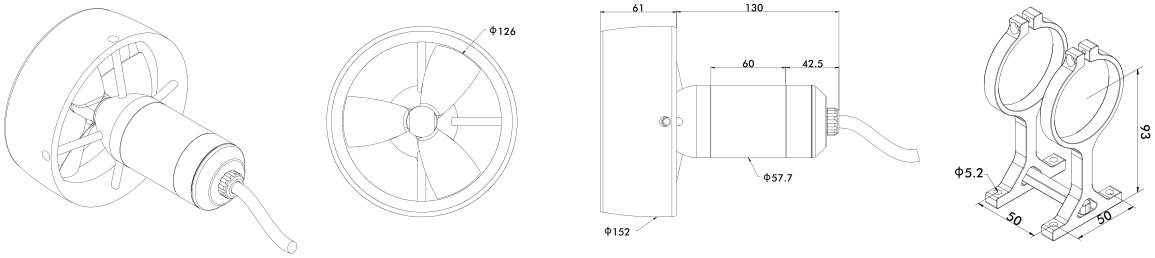
ANALYTICAL GRAPH OF MOTOR OPERATION





DW25

PRODUCT DRAWING

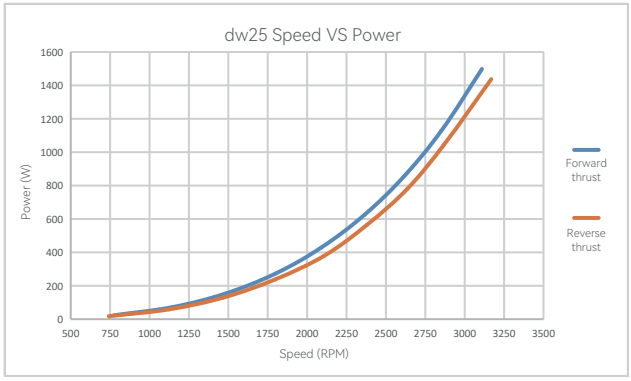


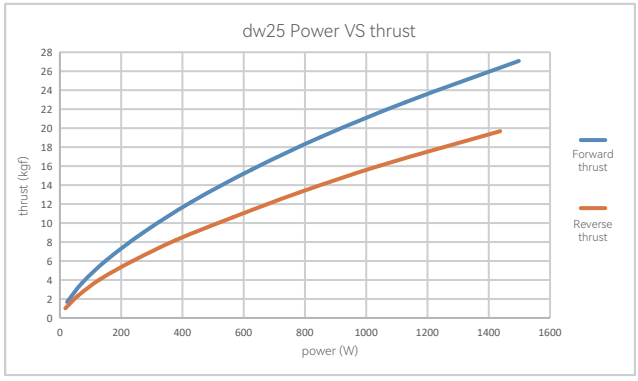
SPECIFICATIONS

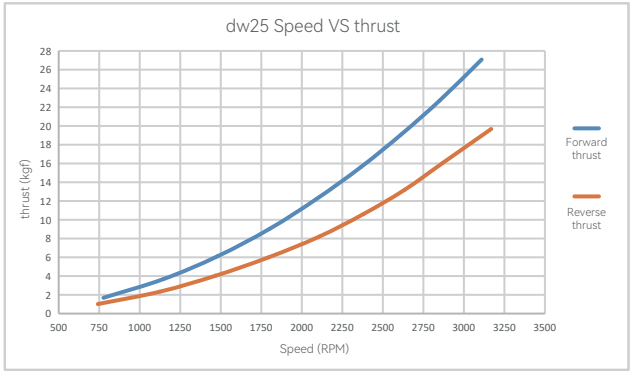
Application	Unmanned Surface Vehicle Electric Surfboard, ROV, Etc
Rated Power	1315W
Rated Voltage	48V
Maximum Mooring Thrust	≥25kgf
Applicable Water Depth	0~350M
Material of The Nozzle	POM
Material of The Shell	Aluminum Alloy
Material of The Propeller	Aluminum Alloy

Propeller Specifications	Ø126mm
Main Shaft Sealing Method	Low-Resistance Dynamic Seal
Main Body Structure	Two-Stage Sealing Chamber
Cable Specifications	TPU (Matte) Ø9.5mm-3*12AWG-1.5M
Driver	External
Weight	1790g (In Air)/995g (Under Water)
Environmental Conditions	0~50℃ (Storage)/0~40℃ (Operation)
Usage Tips	Rinse With Clean Water Promptly After Use

ANALYTICAL GRAPH OF MOTOR OPERATION







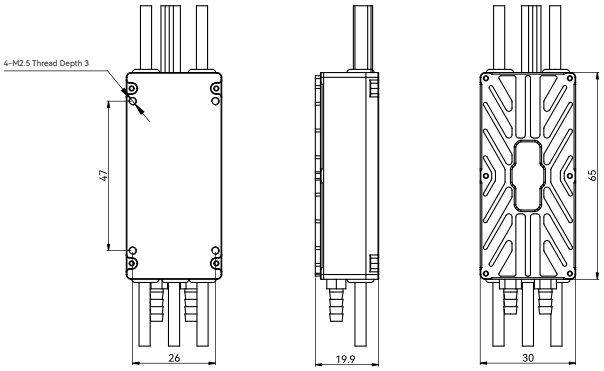
TW Series Underwater ESC

TW40A 12S 98

TW80A 12S 98

TW40A 12S

PRODUCT DRAWING



SPECIFICATIONS

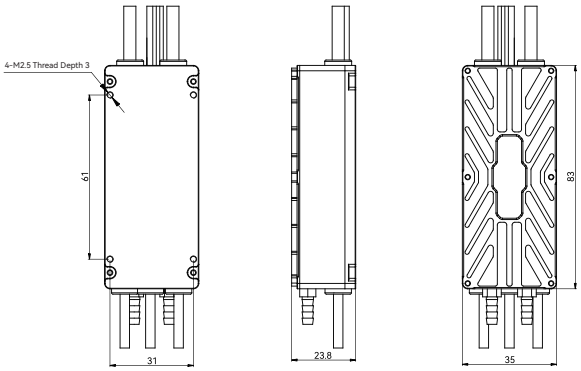
Applicable Models	SW7/SW12/DW10/DW15
Size	78*30*19.9mm
Single Unit Weight	90.33g
Package Dimensions (Includes Dual ESCs)	140*137*77mm
Wiring Harness Description	Power Cable (Input): Red (Positive) Black (Negative) Three-Phase Cable (Output): 3*Black Signal Cable*1 Serial Cable*1

PERFORMANCE SPECIFICATIONS

Input (V)	6~12S
Throttle Range (With Throttle Midpoint At 1500 μs)	1000~2000us
Bidirectional Motor Rotation	1000~1500us Forward/ 1500~2000us Reverse
Communication Mode	Serial

TW80A 12S

PRODUCT DRAWING



SPECIFICATIONS

Applicable Models	SW17/DW20/DW25
Size	83*35*22.8mm
Single Unit Weight	149.33g
Package Dimensions (Includes Dual ESCs)	140*137*77mm
Wiring Harness Description	Power Cable (Input): Red (Positive) Black (Negative) Three-Phase Cable (Output): 3*Black Signal Cable*1 Serial Cable*1

PERFORMANCE SPECIFICATIONS

Input (V)	6~12S
Throttle Range (With Throttle Midpoint at 1500 μs)	1000~2000us
Bidirectional Motor Rotation	1000~1500us Forward/ 1500~2000us Reverse
Communication Mode	Serial

CubeMars
MOTIVATE ADVANCED ROBOTIC SYSTEM

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Official Website

Official Account